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A real-time path planning for reducing vehicles traveling time in cooperative-intelligent transportation systems

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ABSTRACT

Intelligent Transportation Systems (ITS) have the potential to enable efficient real-time path planning applications capable of handling congestion traffic problems using vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communications. However, path planning applications are facing the biggest challenges related to the increase in communication cost caused by redundant communications, which has a destructive impact on the overall network performance. So, the communication design should be carefully adjusted to avoid this problem. In this paper, we propose an effective real-time path planning strategy based on a cooperative information-sharing approach between vehicles and RSUs, which takes into account the communication cost by using a shortest path-based relaying of communications between RSUs to avoid communication redundancy. First, the assigned roles for each item in the network are revised to reduce communication cost, for example, vehicles can share travel time information with RSUs by adopting only single hop communications, especially since each RSU can cover a set of road segments in its surrounding region. Next, we propose a simple shortest path-based routing, which is partially similar to Geographic Source Routing (GSR), but it differs from GSR in the way the vehicles relay communications between each other in road segments as we use the weighted p-persistence broadcasting method to reduce unneeded communication redundancy, while under disconnectivity, a carry-and-forward mechanism is used to allow information sharing with next hop RSUs. Both the proposed path planning strategy and the communication architecture are evaluated according to several scenarios and performance metrics. The simulation results show that our strategy can significantly improve vehicles' traveling time on the road as well as decreasing communication cost.

1. Introduction

The Intelligent Transportation Systems (ITS), which are typically characterized by Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication capabilities, are considered to be the future of VANET technology [1]. ITS have the ability to support a large number of applications. These applications can provide many services to drivers, including real-time warning to improve traffic safety such as collision warning, sharing traffic information to avoid congestion, using GPS-based navigation systems, providing intelligent traffic signaling, cooperative incident management, speed violation detection, and so on [2–4]. In VANET, vehicles can share information about the traffic state with each other via wireless communications, or either with RSUs. In this way, vehicles can exploit this information for collaborating to avoid some challenging events on the road (see Fig. 1). However,

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unexpected traffic congestions are amongst the most challenging problems faced on the road. Traffic congestion appears due to a sudden transition between a free-flow traffic state and a traffic jam, where in free-flow state, vehicles can move very fast, whereas in the case of traffic jam, the traffic state becomes extremely challenging to deal with [5–8]. This makes us think about the factors responsible for the formation of traffic congestions and if there are any indicators to inform us whether traffic congestions are about to occur or not somewhere in our city. Several factors can be responsible for traffic congestion, but the increase in vehicle density is among the well-known key factors, especially when this is combined with the lack of efficiency of the urban infrastructure and its ability to handle the fast growth in the number of vehicles. When the traffic system is constrained by these conditions, it becomes more susceptible to congestion, and therefore at that time, even the smallest acts by drivers can cause massive traffic congestion or a traffic jam. Several actions or factors can cause traffic congestion, including cars stopping on the roadway due to a lack of options for direction change, car accidents, lane closure, construction work, broken traffic signals, imperfect driving styles, traffic interweaves at crossings and roundabouts, and so on [9–11]. When looking at these challenges, it becomes more obvious that ITS with their innovative services are considered to be an attractive solution to develop applications capable of notifying drivers about events on road and the traffic state.

Over the last few years, several contributions dealing with traffic problems based on path planning applications are proposed in the literature [7,12–16]. As traffic congestion problems are at the top list of most encountered problems on the road that have a direct impact on drivers' daily trips, most categories of researchers adopted real-time strategies based on VANET networks and shortest path methods such as Dijkstra algorithm [17], or either based on optimization methods such as ACO [13], improved versions of salp swarm algorithm (SSA) [15], Exponential Bird Swarm Optimization Algorithm (Exp-BSA) [14], stochastic Lyapunov optimization technique [18], and so on.

Some of the researchers focused on the optimization of vehicular traffic under emergency in varying challenging traffic scenarios, for example, to minimize the travel time and prevent accidents on the road [14], or to avoid rescue delays due to traffic congestion in the urban road network [15], or to improve the effect of path planning in emergencies [19]. In the same context, various intelligent strategies that focus on re-routing capabilities in ITS have been proposed as solutions against unforeseeable situations that may occur on the roads [20], either to help drivers make decisions before arriving at such intersections depending on the traffic state [21], or either to improve the situation awareness on the road based on a real-time collection of traffic information [22].

One of ITS' challenges is the communication overhead, which is considered among the main causes responsible of network overhead problem [23,24]. This is due to an increase in the amount of unneeded broadcasts by nodes in the network, and therefore this frequent retransmissions of the same packets by several nodes can cause a rapid increase in communication cost, which finally leads to a worse degradation of the network performance and a high latency [25]. In fact, this last can be a big problem, especially for some critical applications such as path planning and safety-related applications. In some probable scenarios, drivers may find themselves in a dangerous situation that leads to accidents.

However, most of the proposed works focus only on the optimization side of such path planning applications, while neglecting the fact that the effectiveness of such proposed path planning application is significantly affected by the network condition and its ability to handle the increase in communication cost due to a huge amount of information exchange and unneeded redundancy of communications. Consequently, the traffic flow will encounter serious perturbations due to an inaccuracy in the received information about the traffic state as several communications will fail to attend timely their destinations because the channel is congested, and therefore this means an increase in the latency. Here, we should ask ourselves the following question: What will happen in the case of emergencies ? Furthermore, among the most important issues that previous works also neglected is that when such a path planning strategy uses the network resources, there may exist other applications or services sharing the same network resources in parallel with such running application, and therefore this will amplify the communication cost more than we imagine.

Motivated by the limitations of previous research in this trend, we propose an effective real-time path planning strategy based on a distributed and cooperative information sharing mechanism by avoiding communication redundancy and minimizing the communication cost. In other words, we redefine the roles for each item in the network and the relationship between all items to control the communication cost and avoid unended communications, including the way the nodes relay communications between each other. Indeed, the contributions of this work are threefold:

- We have proposed a distributed method to collect traffic information in terms of travel time in road segments to avoid both computation complexity and communication overhead based on a cooperative approach between vehicles, RSUs and the central server. The assigned roles for each item in the network are described as follows:
    i. Each vehicle is able to measure the traveling time in such road segment, and then it transfers this information to a nearby RSU via only one single hop communication to avoid unneeded redundancy of communications, and therefore this helps in avoiding such congestion in the network channel and probable communication overhead.
    ii. RSUs are deployed at intersections and each one can cover a set of road segments in its surrounding region. Each one is able to collect and classify the received information of travel time from vehicles in each nearby road segment, and then the RSUs compute the average travel time per road segment on a specific time window. Finally, all average travel time information are included in one beacon packet to reduce an extra communication cost in the network, and then this beacon is sent to a central server via multi-hop communications at the end of time window, and vice versa this operation is repeated each time a new time window is created.
    iii. In the request and response phases, the server exploits the received information to calculate shortest paths based on Bellman-ford algorithm, and then responds to vehicles' requests with reply packets via multi-hop communications between vehicles and RSUs.

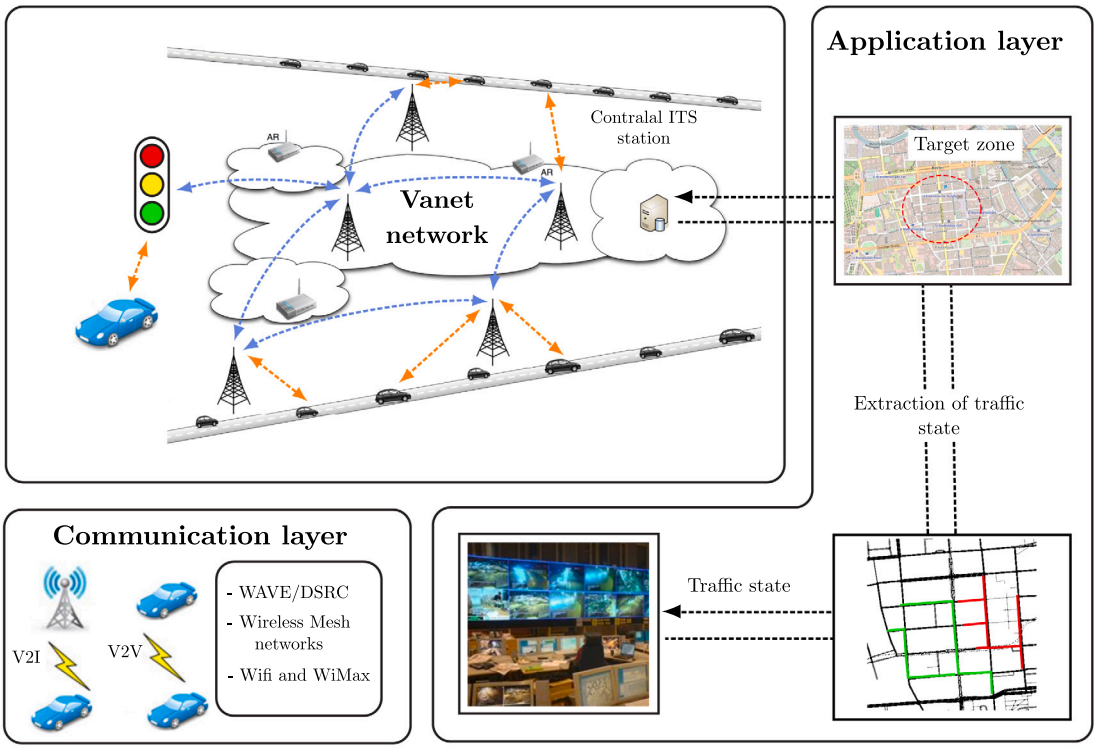


Fig. 1. Illustration example of an intelligent transportation system (ITS).

- To avoid communication overhead caused by redundant communications when RSUs need to share traffic information with the central server or during communication exchange in both request and response phases between vehicles and the central server, we introduce a simple shortest path based routing which is partially similar to Geographic Source Routing (GSR) as it profits of immobility of RSUs and use them as relay nodes. But, our proposed routing method differs from GSR in the way the vehicles relay communications between each other in road segments when such a RSU needs to relay communications to a neighbor RSU (resp. next hop RSU in the shortest path). Due to disconnectivity in some road segments between RSUs, a weighted p-persistence broadcasting method is used to allow vehicles relay communications to next hop RSUs at end of road segments. The adopted probabilistic broadcasting method ensures a oriented communication between vehicles towards the next hop while avoiding unneeded relays, which avoid extra communication cost in the network.
- Finally, we analyze the proposed path planning strategy based on different traffic scenarios and beacon time intervals (BTI) used by the RSUs to share travel time information with the central server. For this purpose, we consider several performance metrics, including travel time, mean speed and total number of travels. Moreover, we analyze the communication cost of our communication architecture in terms of packet delivery ratio, packets volume, communication redundancy, and latency.

The rest of the paper is organized in the following way. We first review the related work in Section Section 2. In section Section 3, our path planning strategy and the adopted approach to minimize redundant communication overhead are introduced, then the communication cost is analyzed. In section Section 4, we describe the considered mobility scenario, simulation environment, network model, and performance metrics. Section Section 5 presents the simulation results and the paper is concluded in Section Section 6.

2. Related work

In this section, we review and discuss some of the existing works in the literature realized over the last few years to overcome the issue of congestion in vehicular traffic systems based on navigation systems and path planning strategies as new emerging services for intelligent transportation systems (ITS). Table 1 summarizes different features of the proposed solutions found in the literature.

The problem of traffic congestion has attracted higher attention from the research community due to its importance and its dependence on many factors that impact the traffic state such as driver behaviors, vehicle density, the quality of infrastructure of the road network, and so on. For example, the authors in [18] proposed a new method to dynamically calculate the current traveling time in a street by using RSUs at the start and the end of each street. The efficiency of the proposed method is investigated in a real city as a VANET scenario based on a simulation framework including SUMO, OMNET++, and Veins. However, the authors did not report how vehicles trigger broadcasts of the calculated travel time when they reach the end of the street. In [18], the

Table 1
Comparison among some path planning strategies in the literature. (–) means that communication architecture is not provided.

Arch.	Ref.	Net. issues			Traffic view			Features			
		Broadcast storm	On-demand information	Overload control	Local	Global	Dynamic	Traffic jams prediction	Real-time re-routing	Periodic re-routing	Region-based
V2V & V2I	Our		✓	✓		✓	✓		✓		
–	[26]					✓			✓		
V2V & V2I	[12]	✓		✓		✓		✓	✓	✓	
–	[27]		✓			✓				✓	
V2V & V2I	[26]	✓	✓			✓	✓		✓		
CNI	[21]	✓		✓		✓		✓	✓		
V2V, V2I & P2I	[20]	✓	✓			✓			✓		
V2V, V2I & I2I	[21]	✓		✓		✓			✓		
V2V, V2U & V2BS	[22]		✓	✓		✓		✓	✓		
–	[19]					✓		✓			✓

authors proposed a real-time path planning algorithm, which not only improves the overall spatial utilization of a road network, but also reduces average vehicle travel cost and prevents vehicles from getting stuck in congestion as well. The proposed algorithm is based on a stochastic Lyapunov optimization technique that is exploited to address the globally optimal path planning problem. The results show that the proposed path planning algorithm outperforms traditional distributed path planning in terms of balancing spatial utilization and drivers' travel cost. In [28], the authors introduced a novel dynamic vehicular path planning solution. The proposed solution does not relay on infrastructures to collect traffic information. Instead, it utilizes a density-speed traffic flow model to predict traffic conditions. Also, a dynamic candidate path selection algorithm is developed to reduce the redundant data collection overhead. This approach has been extensively evaluated using large-scale traffic trace-based simulations. The obtained results show that the proposed approach outperforms some existing solutions in terms of communication efficiency and path planning effectiveness. In [29], the authors proposed a route planning algorithm to calculate the route with the shortest traveling time based on real-time traffic information by using VANET. At each road segment, a traveling vehicle can exchange information about traffic state with other neighboring vehicles by using the IEEE 802.11p standard and a GPS navigation application implemented on the Android platform. This approach uses the average recorded driving speed as an indicator of traffic state at road segments. Another class of approaches is based on using path planning applications to reduce road traffic congestion and avoid traffic jams in large cities. For example, in [30], to avoid traffic jams in large cities, the authors proposed an intelligent traffic system called CIDMERA, which improves the overall spatial utilization of the road network. CIDMERA can also reduce average vehicle travel costs by preventing vehicles from getting stuck in traffic. In [31], the authors proposed a distributed and collaborative traffic congestion detection and dissemination system that uses VANET. This system is based on the use of a smartphone by each driver to detect location through GPS. This information is sent to a remote server that detects traffic congestion. Once congestion is confirmed, the congestion information is disseminated to the end-user phone through RSUs. In [21], to determine optimal routes by minimizing costs like travel time to their destination, especially for vehicles whose original route is adversely impacted by an unexpected severe road event, authors proposed a strategy based on re-routing of vehicles. This last is enabled just before upcoming road intersections as convenable areas to make decisions based on the received information from a central server about the estimation of the remaining lifetime of the road event. In [22], to avoid some difficulties observed in navigation systems related mainly to a lack of local knowledge of the area of navigation solutions and inflexibility against unforeseeable situations that may occur on the roads, especially under increasing vehicle density and the growth of accidents in urban areas. The authors proposed a novel three-tier architecture, called SDN-enabled Approach for Vehicle Path-Planning (SEARCH) to improve the situation awareness on the road and provide an efficient real-time collection of traffic information, and then help drivers to make the correct decisions. The proposed architecture combines multiple technologies, including Unmanned Aerial Vehicles (UAVs), Vehicular Ad hoc Networks (VANETs), 5G based cellular systems, and Software-Defined Networking (SDN) to provide a fast exchange of information about the traffic state. Accordingly, several parameters are used to facilitate the interaction between both vehicles and the driving environment, and therefore to achieve a fast re-routing of vehicles. In [13], the authors proposed a new distributed strategy to optimize traffic road congestion in real-time based on VANET networks and the Ant Colony Optimization (ACO) method. For this purpose, the VANET network is used to facilitate the exchange of traffic information between vehicles and RSUs, and then communicate the shortest paths to drivers. In [32], the authors aim to improve navigation in modern traffic systems and allow drivers to select the best paths and then reduce travel costs. Accordingly,

Table 2
Table of used mathematical symbols.

Symbol	Description
$\mathbb{V}$	The set of vehicles in road the network, where $\mathbb{V} = \{v_1, v_2, v_3, \dots, v_n\}$
$\mathbb{I}$	The set of intersections, where $\mathbb{I} = \{I_1, I_2, I_2, \dots, I_m\}$
$\mathbb{S}$	The set of all road segments, where $\mathbb{S} = \{rs_{i,j}, i \neq j\}$
$\mathbb{R}$	The set of all RSUs in the network, where $\mathbb{R} = \{R_1, R_2, R_3, \dots, R_m\}$
$\mathbb{P}$	The set of all packets in the network, $\mathbb{P} = \{pk_1, pk_2, pk_3, \dots, pk_{n'}\}$
$\mathbb{B}$	The set of all beacons in the network, $\mathbb{B} = \{b_1, b_2, b_3, \dots, b_q\}$
$\mathbb{T}$	Set of weights corresponding the average travel time per road segment $rs_{i,j}$, where $\mathbb{T} = \{T_{i,j}, i \neq j\}$ and $T_{i,j} \rightarrow R_+^*$
$n_{i,j}$	The number of vehicles in the road segment $(rs_{i,j})$, where $i \neq j$
$Sp_{i,j}$	GPS reference point that indicates the start of road segment $rs_{i,j}$
$Ep_{i,j}$	GPS reference point that indicates the end of road segment $rs_{i,j}$
$St^u_{i,j}$	The time when vehicle v_u starts moving on such road segment $rs_{i,j}$
$Et^u_{i,j}$	The time when vehicle v_u reaches the end of road segment $rs_{i,j}$
DB_m	A locat database that stores the list of road segments information at the level of each RSU R_m, including their corresponding IDs and the average travel time
DB_{ser}	A central database that stores road segments related information at the level of server and their corresponding IDs
BTI	The beacon time interval to send the average travel time by RSUs to the server
BST	The beacon send time of average travel time by RSUs to the server
E_0	Is the precision error used to detect vehicle at a specific position in road segment $rs_{i,j}$
I^{id}_{cur-s}	Is the id of current source intersection
I^{id}_{prev-s}	Is the id of previous source intersection
I^{id}_{dest}	Is the id of destination intersection

they propose a hybrid architecture that uses a multi-hop routing algorithm to permit vehicles to enlarge their driving performance and road safety. The proposed routing algorithm is combined with an intelligent transportation system (Hybrid VMR-ITS) to allow efficient instantaneous communications between vehicles, roadside units (RSUs), and vehicle-traffic servers. In the same context, in [17], to avoid time spent at intersections, the authors studied the path planning problem in the context of car navigation systems by considering intersection attributes. They handled this problem as an optimization problem based on a deterministic network. For this purpose, first, the authors constructed a mathematical model for studying the problem, and then to solve it, they proposed a reverse labeling Dijkstra algorithm (RLDA) based on the traditional Dijkstra algorithm. Finally, the authors proved theoretically that the RLDA algorithm has a lower polynomial time complexity and conducted extensive experiments to compare it in terms of running time with the following algorithms: PSO, GA, ACO, NNA, and OPABRL.

Other categories of contributions focus on re-routing capabilities as an important aspect of intelligent transportation systems (ITS), which are capable of providing more flexibility of traffic flow for drivers when the matter is related to the need to avoid traffic congestions or unexpected traffic jams. In [26], vehicle route planning was proposed to optimize vehicle route and speed simultaneously using both traffic data and vehicle characteristics to improve fuel economy for a given expected trip time. For this purpose, a simulation model combining a traffic model based on SUMO (Simulation of Urban Mobility) with a Simulink powertrain model, is developed to validate the proposed method. In [12], another vehicle route planning based on an efficient real-time information-sharing mechanism is proposed to achieve real-time traveling time estimation and dynamic path planning. It uses a sharing mechanism based on a distributed transportation system with RSUs, which has lower computing complexity and less redundancy. This method is evaluated With VanetMobiSim, a simulator at both macroscopic and microscopic levels. In [27], an automatic vehicle re-routing mechanism is proposed based on graph theory concepts to re-compute routes under specific traffic conditions (e.g., traffic jams). Based on the SUMO simulator and previous constraints, realistic mobility is generated based on a real map, where vehicles are allowed to change their routes during their trips dynamically and periodically according to the current traffic state (see Table 2).

Other studies analyze the problem of traffic congestion with different approaches and under some constraints like energy awareness and connectivity, for example, in [20], the authors proposed a VANET-based A*(VBA*) route-planning approach based on the real-time exchange of traffic information with nearby vehicles via the vehicular ad hoc network (VANET) to allow vehicle re-calculate their paths to the destination to save time and fuel. In the same context, in [33], the authors aim to optimize energy utilization due to the limitations of electric vehicle (EV) battery technology and relevant supporting facilities. They highlighted the risk of the breakdown of EVs during driving, which will have a worse effect on the traveling experience of EVs. To overcome this problem, they proposed a method to facilitate the travel of EVs based on a path-planning strategy by considering EV power supply. First, based on real-time traffic state, a dynamic energy model of EVs was established considering the driving energy and accessory energy. Second, a multi-objective travel path planning model for EVs was constructed considering the power supply, traveled distance, time, energy, and charging cost as the optimization objectives. In [34], to provide Connected and Automated Vehicles (CAVs) driving on a lane-free highway, the authors introduced a path-planning algorithm based on a novel paradigm for vehicular traffic in the era of CAVs. Accordingly, the adopted approach uses a simple model of vehicle kinematics by taking into account the appropriate constraints for control variables and road boundaries. To this end, a set of competitive sub-objectives are designed to provide efficient vehicle advancement and allow vehicles to avoid both collisions between each other and unsuitable maneuvers. Finally, this conducted authors formulate two main elements, including a nonlinear Optimal Control Problem (OCP) for each ego

vehicle, and a Feasible Direction Algorithm (FDA), which is employed for its computationally to provide an efficient numerical solution. In the same context, to analyze the impact of vehicle connectivity on the efficiency of advancement of automated vehicles (AVs) and the efficiency of the emerging traffic flow. The authors in [35] proposed a path-planning algorithm for connected and non-connected automated road vehicles on multilane motorways derived from the opportune formulation of an optimal control problem. Accordingly, the objective function to be minimized contains appropriate specifications related to vehicle advancement, passenger comfort, and avoidance of collisions with other vehicles and road departures. To provide a real-time exchange of information about their last generated short-term path, the connected vehicles can use (V2V) communications. To optimize the numerical solution, two main algorithms are used, including both the Feasible Direction Algorithm (FDA) and a simplified Dynamic Programming (DP) algorithm. Finally, the path-planning algorithm was evaluated based on multi-lane motorways characterized by mixed traffic, while vehicles are driven at different penetration rates. In [16], the authors proposed a real-time path planning for large-scale scenarios by taking into account the regularity of people's work and the data history of traffic flow to provide an efficient prediction of future congestion scenarios in such regions in the road network. For this purpose, a clustered queue communication mechanism and queue-based shunting are used. In the same context, to solve the problem related to the uncertainty of prediction information in traditional path planning algorithms, in [36], the authors proposed a path planning algorithm based on Mutual Assistance Trip (MAP), which uses the concept of mutual assistance in travel. The proposed algorithm aims at minimizing a MAP-based path planning objective function in terms of travel time. Further to the objective function, another function is introduced to mix links with certain penalties via two main coefficients: an obedience coefficient μ and an intensity coefficient of mutual assistance information θ. Finally, by using both the greedy policy and the roulette mechanism, the path planning algorithm can select the convenient travel direction.

Due to the importance of traveling delay in the case of critical situations such as emergencies, especially with the exponential growth of traffic accidents, in [14], the authors proposed a dynamic path planning to provide an efficient routing under emergency by avoiding congestion, minimizing the travel time, and preventing accidents. The proposed path planning method includes two main techniques, namely travel-time forecasting and bi-model routing. The travel time is predicted using the Support Vector Regression model and the bimodal is employed to provide optimal routing of emergency vehicles. The bimodal consists of two steps, including K-path discovery and an optimal path selection process. To compute the shortest paths, the authors adopt the Exponential Bird Swarm Optimization Algorithm (Exp-BSA). In [19], the authors aim to improve the effect of path planning in emergencies by proposing a novel trajectory data cube (TDC) to model, analyze and process trajectory data. For this purpose, three trajectory hierarchical trace-back algorithms (HTB-p, HTB-v, and HTB-KF) are proposed based on the TDC model to restore the missing information in the trajectory, and then restore the traffic flow of the missing trajectory. In the same context, to avoid rescue delays due to traffic congestion in the urban road network, the authors in [15] proposed a real-time traffic control with congestion index constraints in emergency vehicle dispatching and a two-stage optimization model and algorithm. In the first stage, both the salp swarm algorithm (SSA) and the Dijkstra algorithm are combined to form a novel hybrid algorithm designed to get multiple alternative paths. In the second stage, an improved salp swarm algorithm (ISSA) was proposed to obtain the best evacuation schemes and the optimal rescue paths for emergency vehicles. Indeed, the introduced path planning strategy was evaluated according to the travel time of the alternative path before and after evacuation.

In reference to the relevant works discussed above, we conclude that the attention has been highly devoted to the path planning problem as an optimization problem, while we still lack to give more attention to communication overhead problem due to an explosion in communications redundancy between all participating items in the network (i.e., vehicles, RSUs, and central communication units). In fact, both the adopted communication design and the assigned roles for each item in the network should be revised to avoid such undesirable communication exchange and redundant communications of the same information, which will reduce the network reliability, especially since the network should simultaneously provide other services based on the same V2V and V2I architecture.

In this work, our goal is to fill this gap by proposing an efficient real-time path planning strategy based on a cooperative and distributed approach to facilitate the collection of traffic state information by introducing a robust communication architecture capable of avoiding redundant communications in the network. To this end, the proposed path planning strategy can provide a cooperative exchange of real-time information about the traffic state by reducing the communications cost between items in the network, including vehicles, RSUs, and the central server deployed outside of the road network near to its the right side. To avoid the redundancy of communications, and then decrease the network overhead, we first redefine the assigned roles for each item in the network and the way they communicate with each other. Next, we introduced a simple shortest path-based routing which is partially similar to Geographic Source Routing (GSR), but it differs from GSR in the way the vehicles relay communications between vehicles in road segments when such RSU needs to communicate with a neighbor RSU (next hop RSU). Here, we take care of both communication cost and the communications' direction to avoid extra further communications redundancy. In fact, such forwarding of communications to an opposite direction to the next hop is a worse problem which will increase unneeded relays of the same communication, and therefore an overall degradation of network performance becomes more probable to occur. To avoid this scenario, we introduced a probabilistic broadcasting method that allows vehicles in such road segment to relay communications to a next hop RSU at the end of this road segment, and then provides connectivity between successive RSUs in each assigned shortest path to route and deliver communications to a specific destination. This is expected to limit a huge redundancy of communications, either during the periodic sharing of traffic information by RSUs with the central server, or either during communication exchange in both request and reply phases between vehicles and the central server, where both vehicles and RSUs are considered relay nodes.

In fact, the most attractive features of our proposed path planning strategy reside in its ability to redefine the assigned roles for each item in the network and its ability to avoid redundant communications at different levels of communication. This communication architecture includes also a decentralized collection mechanism of traffic information, which limits the number

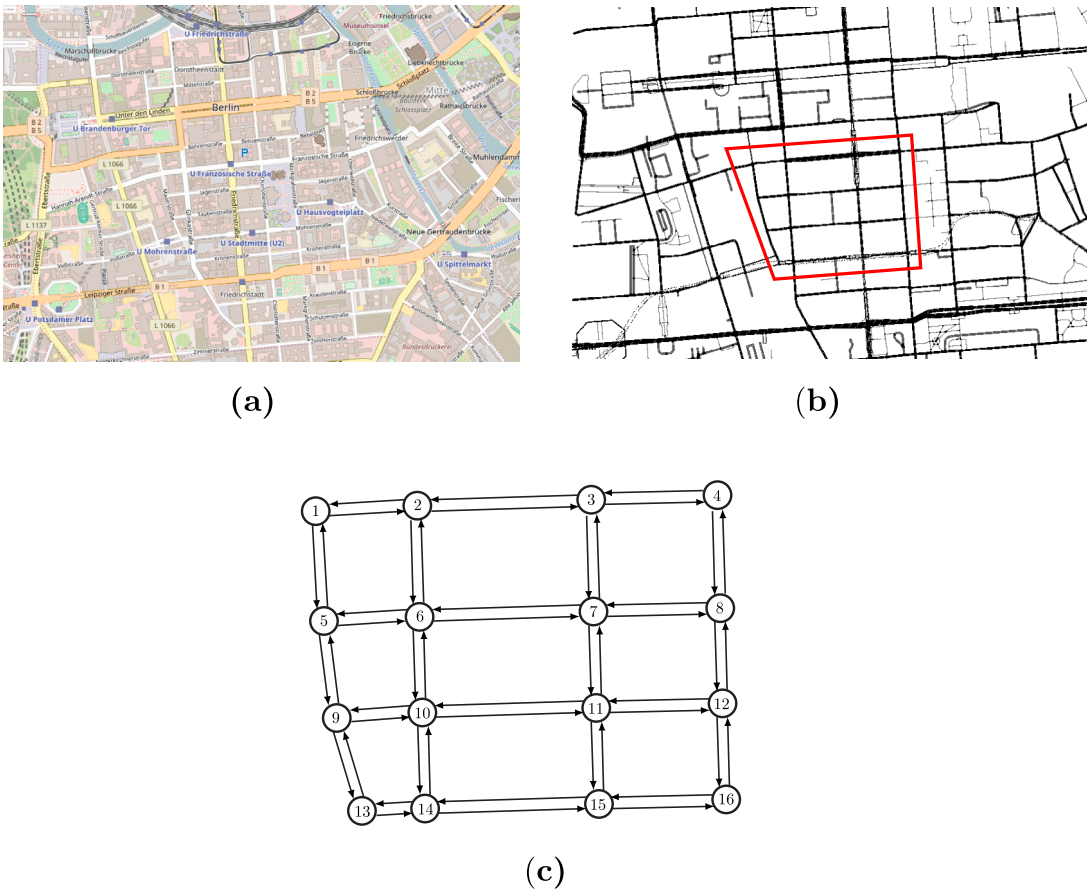


Fig. 2. Illustration of the considered traffic scenario: (a) Berlin map, (b) The simulation area and (b) The directed weighted graph extracted from the map.

of items participating in this operation, and then limits extra communication cost. All these improvements will allow drivers to profit from the high reliability of such path planning applications during their traveling to preferred destinations and sure they will be confident and motivated to travel under varying traffic conditions (see Figs. 2 and 4).

3. Model descriptions

In this section, we introduce a detailed description of the proposed path-planning strategy. Accordingly, we first give the system assumptions and the model of V2V and V2I communications in the network, and then we describe the interaction between all items in the communication architecture and the way they communicate with each other. In the last part of this section, we introduce our solution to avoid redundant communication problem based on an efficient control approach, and then we realize an analysis of communication cost.

3.1. V2V and V2I communication architecture and assumptions

In this paper, we mainly discuss the role of V2V and V2I communications to collect and share traffic state with a centralized server deployed outside of the road network near its right side (see Fig. 3). Based on the illustration in Fig. 3, at each intersection identified by a pair (i, j) in the road network, we deploy a RSU R_m to facilitate V2V communications among vehicles and V2I communication between vehicles, RSUs, and the server. Accordingly, RSUs are pre-located further to intersections and each one covers a set of road segments to collect travel time from incoming traffic towards the intersection. The location of RSUs helps each RSU R_m to profit from its position which is near an intersection to record the vehicles' travel time and the ID of corresponding road segment $rs_{i,j}$ sent by vehicles that reach the intersection. In fact, the calculation of the travel time based on both the vehicle speed and the length of road segment $rs_{i,j}$ is not considered a good solution as vehicle speed is not constant because it is influenced by real-time traffic density and traffic state. Thus, vehicles should first perform a decentralized measurement of travel time based on the difference in time by using the following steps:

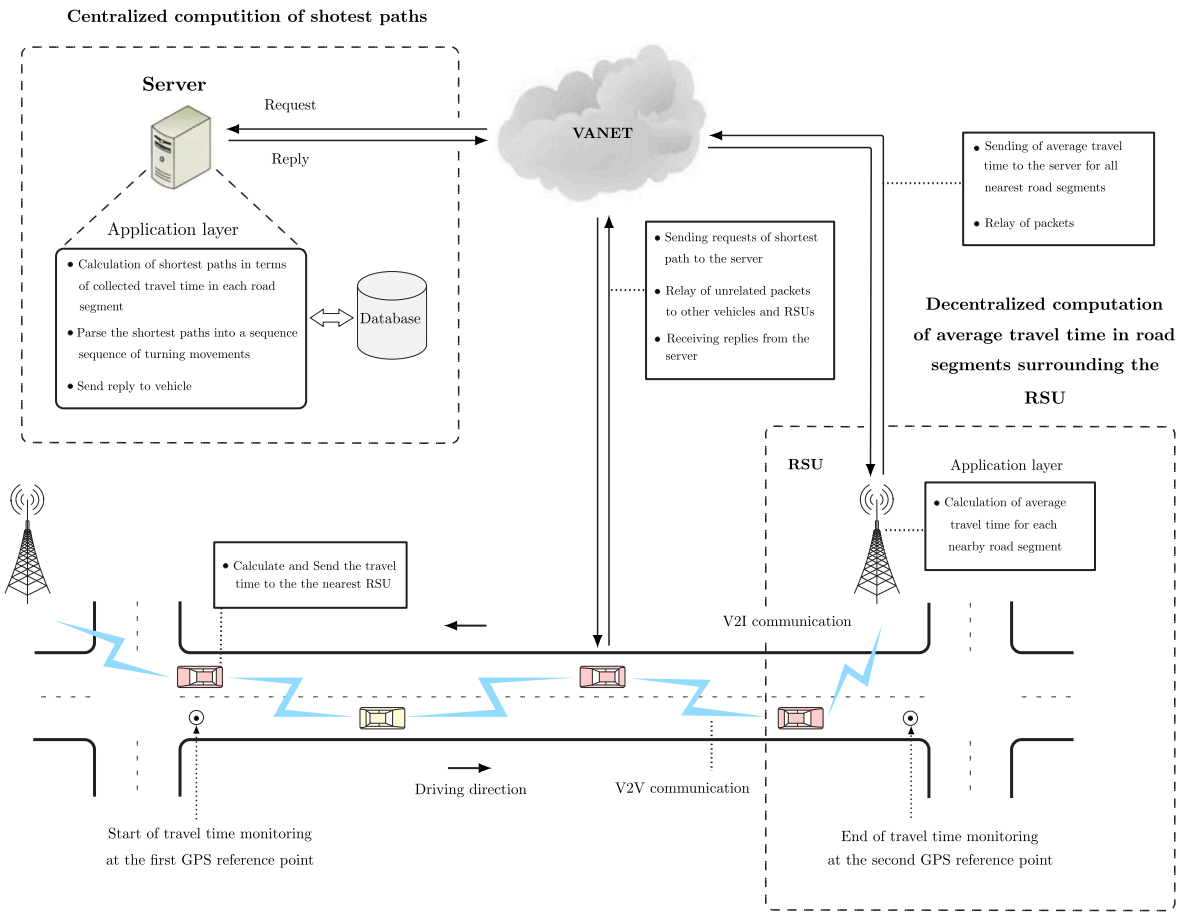


Fig. 3. Illustration of the interaction between all items in the VANET environment.

- A vehicle v_u should be equipped with a GPS system to handle positions in the road network.
- The vehicle v_u should have a pre-stored database of all road segments with the corresponding start and end GPS reference positions ($Sp_{i,j}$ and $Ep_{i,j}$) for each road segment $rs_{i,j}$, where $Sp_{i,j}$ and $Ep_{i,j}$ indicate the start and end positions of road segment $rs_{i,j}$.
- To measure the travel time at road segment $rs_{i,j}$, vehicle v_u should first detect the position $Sp_{i,j}$ with a specific precision error E_0, and then records the corresponding time in the variable $St^u_{i,j}$ (the start time of travel on road segment $rs_{i,j}$) as follows:

$$St^u_{i,j} = t \Longrightarrow (\| Sp_{i,j} - p_u(t) \| \leq E_0) \tag{1}$$

  where $p_u(t)$ indicates the current position of vehicle v_u at time t in road segment $rs_{i,j}$.
- Before arriving at the end of the road segment, vehicle v_u should periodically verify if the end of road segment $rs_{i,j}$ is reached based on the measurement of the distance between the vehicle's position p_u and position $Ep_{i,j}$ with a precision error of E_0, and then we record the corresponding arrival time in the variable $Et^u_{i,j}$ as follows:

$$Et^u_{i,j} = t \Longrightarrow (\| Ep_{i,j} - p_u(t) \| \leq E_0) \tag{2}$$

- When the vehicle u arrives at the end of the road segment, both $St^u_{i,j}$ and $Et^u_{i,j}$ should be already calculated by the vehicle v_u, and then the vehicle v_u should calculate the travel time on this road segment as follows:

$$T^u_{i,j} = (Et^u_{i,j} - St^u_{i,j}) \Longrightarrow (Et^u_{i,j} > St^u_{i,j}) \tag{3}$$

- When calculating the travel time on this road segment $rs_{i,j}$, the vehicle v_u sends a packet containing both the calculated travel time and the ID of corresponding road segment $rs_{i,j}$ to the nearest RSU R_m deployed at the intersection.

The above rules will help vehicles to compute the travel time on road segments and communicate easily this information to nearby RSUs without the need for relay nodes as RSUs are deployed at intersections and each one covers a surrounding region that

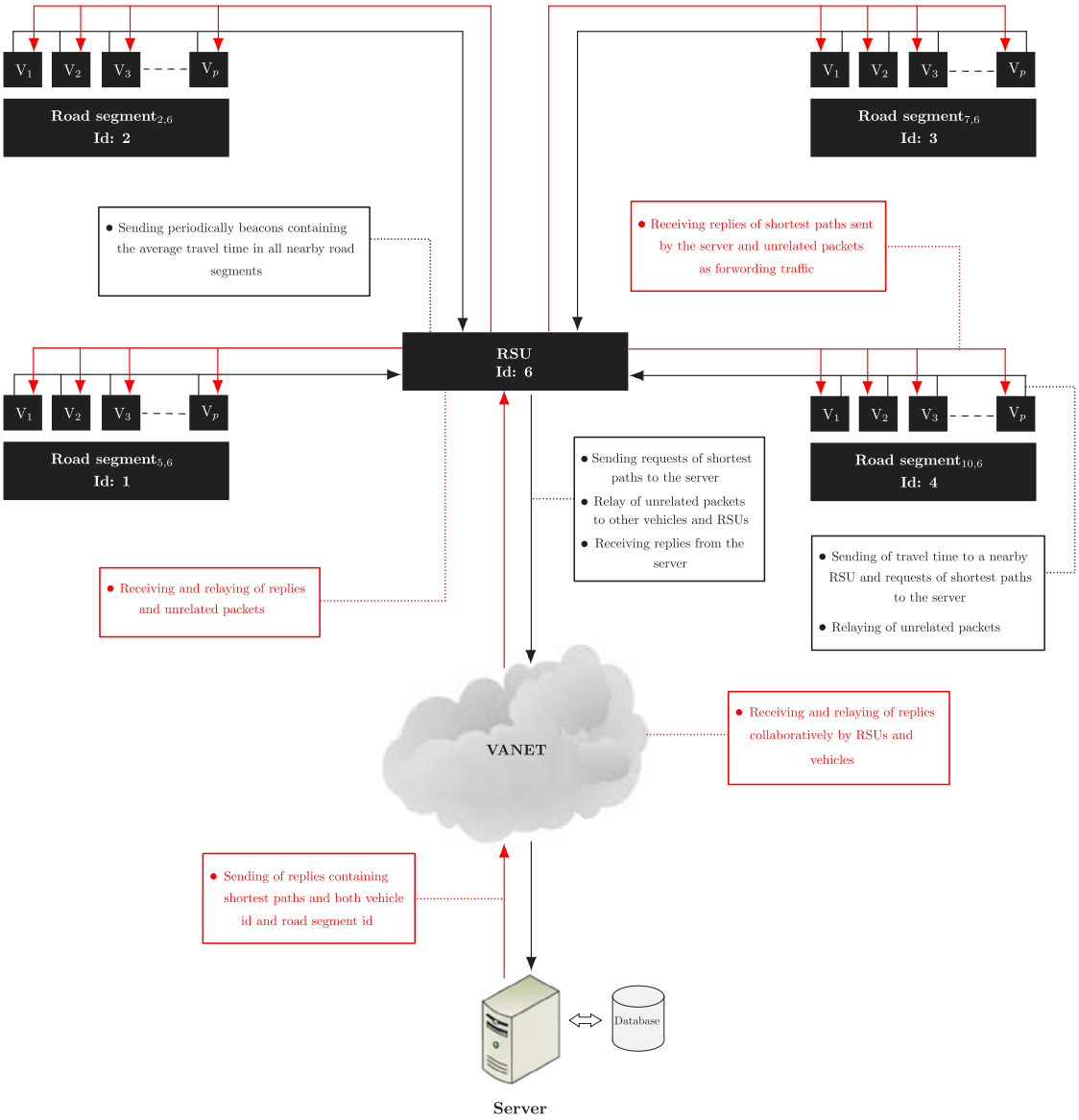


Fig. 4. Packets flow in the network.

contains a set of road segments. The direct exchange of travel time information between vehicles and RSUs is expected to reduce an important amount of redundant communications. Afterward, the transmitted travel time information by vehicles to RSUs should be processed according to the following steps before being sent to the central server:

- Each RSU R_m should classify the received travel time information per road segment $rs_{i,j}$ using the corresponding ID of $rs_{i,j}$ in each received packet and incrementing the variable $n_{i,j}$, which indicates the number of vehicles/packets that the RSU has received per road segment.
- At each time window of the same length as BTI, the R_m should calculate the average travel time per road segment $rs_{i,j}$ in this time window by using Eq. (6) and based on the following sliding window technique (see Fig. 5), and then it stores the calculated average travel time per road segment for all road segments into a local database DB_m with a set of entries (ID of road segment, travel time per road segment, and average travel time per road segment).
- If there is no traffic flow in such a road segment during the time window, the travel time is calculated as follows:

$$\mathbf{T}_{i,j} = \frac{D_{i,j}}{s^{i,j}_{max}} \tag{4}$$

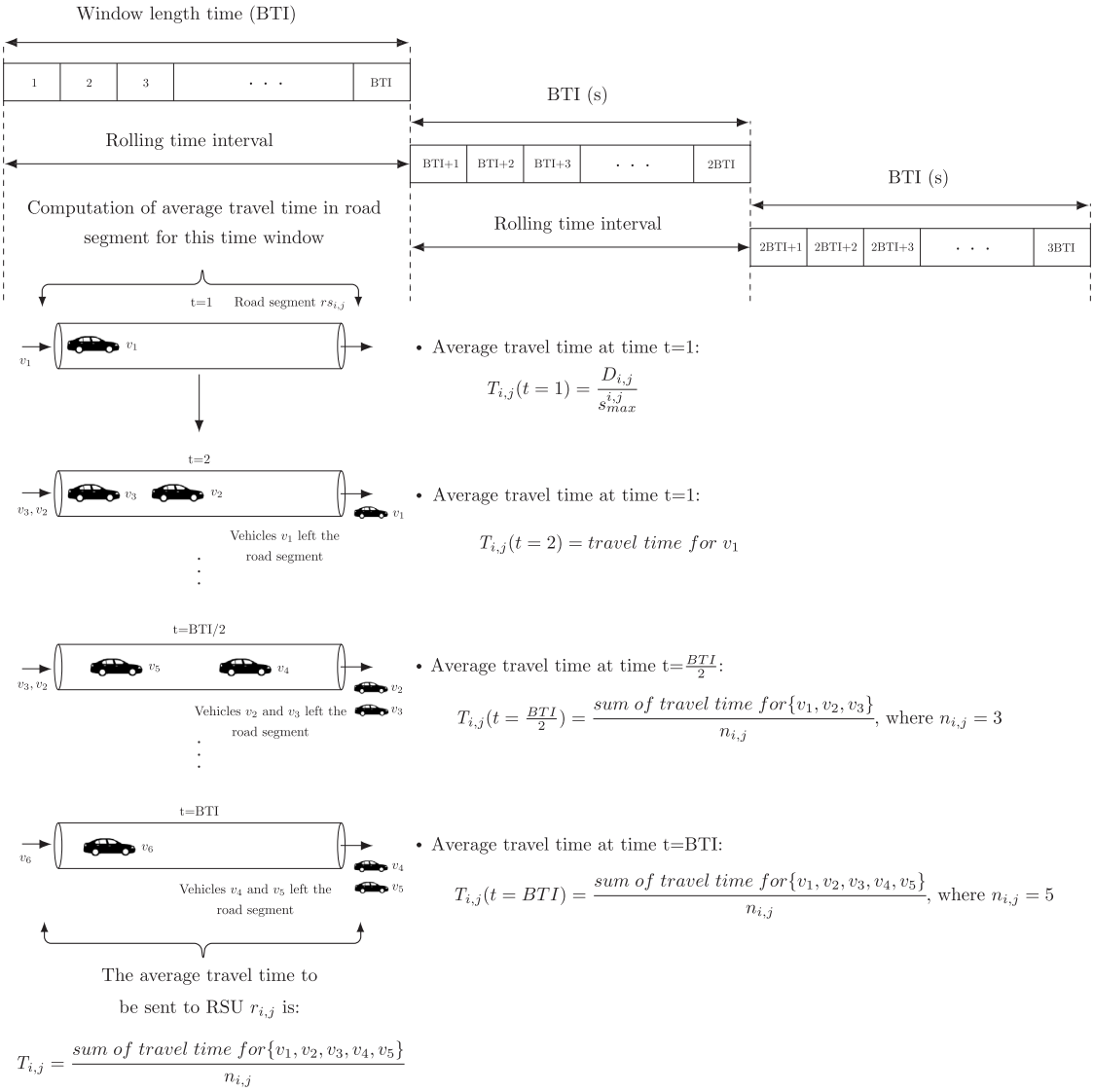


Fig. 5. Computation of average travel time by RSUs based on the sliding time windows technique.

- Otherwise, the average travel time is calculated as:

$$\mathbf{T}_{i,j} = \frac{\sum_{u=1}^{n_{i,j}} \left(Et^{u}_{i,j} - St^{u}_{i,j} \right)}{n_{i,j}} \tag{5}$$

- When the time window is elapsed, each RSU should send a beacon packet that contains the average travel time per road segment $rs_{i,j}$ combined with the corresponding ID of $rs_{i,j}$ (for all nearby road segments to this RSU). The assignment of IDs to road segments will allow the server easily to identify each road segment and link it with its corresponding average travel time when it stores the received traffic information into the database DB_m (see Fig. 4). This packet is sent to the central server via multi-hop communications between both RSUs and vehicles as relay nodes.

3.2. The shortest path method and Vehicle-to-Server communications

In this subsection, we first introduce a detailed description of the way the server process the received average travel time information and how it will exploit this information as weights of edges in graph G, and then it will process the shortest path

method. Next, we describe the exchange mechanism of requests and replies between vehicles and the central server via multi-hop communications. All these tasks are described as follows:

- The server first records the received average travel time information into a database DB_{ser} to facilitate the update of travel time for each road segment based on its corresponding ID inside the received beacon packet sent by such RSU.
- After storing the received average travel time information in the database, the server combines the received travel time information with a graph $G(\mathbb{I}, \mathbb{T}(t))$ as weights of edges, where $\mathbb{I}$ is the set of intersections I_m which represent the set of vertices and $\mathbb{T}(t)$ is the set of weights representing the average travel time per road segment at a specific time window. Based on the graph G, the server process the Bellman-Ford's shortest path algorithm [37] due to its high standing as it is widely used to solve several problems including network routing and optimal route planning in navigation systems. Accordingly, we consider the travel time instead of the distance between each pair of vertices (resp. intersections) according to the traffic state in the road network, where edges represent the average travel time in road segments $rs_{i,j}$ as follows:

$$\boldsymbol{T}_{i,j} = \begin{cases} \frac{D_{i,j}}{s_{max}^{i,j}}, & \text{if } n_{i,j} = 0, \\ \frac{\sum_{u=1}^{n_{i,j}} \left(Et_{i,j}^{u} - St_{i,j}^{u} \right)}{n_{i,j}}, & \text{if } n_{i,j} > 0. \end{cases} \tag{6}$$

  The average travel time is considered an indicator of the current traffic state in each road segment $rs_{i,j}$, which will be used to avoid traffic problems (i.e., congestion and jams). If the average travel time in such a road segment is higher than other road segments, then this road segment will be ignored when selecting the shortest path with decreased travel time. By using the Bellman-Ford algorithm and the received travel time information in road segments, which is considered as an indicator of the current traffic state in the road network, the server can process Bellman-Ford's shortest path algorithm based on the following equation:

$$\boldsymbol{T}_{i,j} = \min_{k \in intersections} \{\boldsymbol{T}_{i,k} + \boldsymbol{T}_{k,j}\} \tag{7}$$

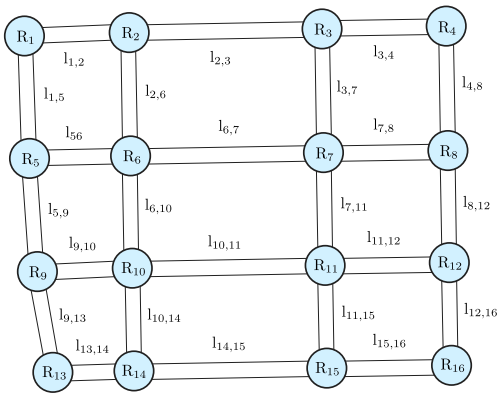
  where $\mathbf{T}_{i,k}$ and $T_{k,j}$ are the travel time for road segments $rs_{i,k}$ and $rs_{k,j}$, respectively.
- Afterward, the server pass to listening mode and starts waiting for vehicles' requests that will be received via multi-hop communications (i.e., V2V and V2I).
- To provide a periodic update of traffic information, at the end of each time interval of length BTI, the server should update the information in the local database DB_{ser} based on the recently received information from the RSUs for this time interval, and then it updates the state of graph G to reflect the current traffic state.

After the above steps are processed by the server, vehicles can interact with the server via multi-hop communications as follows:

- Each vehicle v_u can send a request of a new shortest path during its travel in such road segment $rs_{i,j}$ at the beginning of travel on this road segment and each time it moved to a new road segment before it reaches the intersection. Next, the vehicle's request is forwarded to central server by both RSUs and vehicles as relay nodes, while the sender vehicle of request starts waiting for a reply from the central server to be sent via multi-hop communications based on relay nodes.
- Next, when the vehicle's request is received by the server, the server process the Bellman-Ford algorithm, and then it generates a shortest path for this vehicle as a set of turning movements to help the driver to reach its final destination in the best timing. Afterward, the new shortest path is included in the reply packet, and then it is sent to the vehicle owner of the request via both RSUs and vehicles as relay nodes.
- Finally, when the reply is received by the vehicle source of the request, the vehicle updates its current path if the new received the shortest path differs from the previous one. In fact, the periodic update of vehicle trajectory based on traffic information feedback from the central server is expected to improve significantly the traveling time for drivers and helps them avoid such congestions and other events on the road (see Fig. 11).

3.3. Redundant communications control

In this subsection, we describe our strategy to avoid redundant communication related to the exchange of traffic information between items in the network, especially when performing multi-hop communications to provide the periodic exchange of traffic information between RSUs and the server as well as in both request and reply phases. Accordingly, to avoid redundant communications in the network due to unneeded rebroadcasts and unneeded communications exchange with neighbors, and then avoid channel congestion and long latency, we first profit from the immobility feature of RSUs and try to exploit this advantage to route packets between items in the network with a few numbers of relays. Our idea is to handle RSUs as gateways as well as to perform an efficient control of communications' direction between items in the network. For this purpose, we introduce a simple shortest path-based routing partially similar to Geographic Source Routing (GSR) [38], but it differs from GSR by performing a probabilistic broadcasting method between vehicles in such road segment when a data packet reaches an RSU and it needs to be forwarded to a next hop RSU located at the end of this road segment. Based on the distance between RSUs (weights), the Bellman-Ford algorithm is used to find the shortest paths toward the central server at the beginning of the simulation. Each RSU computes



(a)

	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8	R_9	$\cdots$	R_{16}
R_1	∞	$l_{1,2}$	∞	∞	$l_{1,5}$	∞	∞	∞	∞	...	∞
R_2	$l_{1,2}$	∞	$l_{2,3}$	∞	∞	$l_{2,6}$	∞	∞	∞	...	∞
R_3	∞	$l_{2,3}$	∞	$l_{3,4}$	∞	∞	$l_{3,7}$	∞	∞	...	∞
R_4	∞	∞	$l_{3,4}$	∞	∞	∞	∞	$l_{4,8}$	∞	...	∞
R_5	$l_{1,5}$	∞	∞	∞	∞	$l_{5,6}$	∞	∞	$l_{5,9}$	...	∞
R_6	∞	$l_{2,6}$	∞	∞	$l_{5,6}$	∞	$l_{6,7}$	∞	∞	...	∞
R_7	∞	∞	$l_{3,7}$	∞	∞	$l_{6,7}$	∞	$l_{7,8}$	∞	...	∞
R_8	∞	∞	∞	$l_{4,8}$	∞	∞	$l_{7,8}$	∞	∞	...	∞
R_9	∞	∞	∞	∞	$l_{5,9}$	∞	∞	∞	∞	...	∞
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
R_{16}	∞	∞	∞	∞	∞	∞	∞	∞	∞	...	∞

(b)

Fig. 6. Adjacency matrix of RSUs based on the considered road network.

its shortest path toward the central server and keeps only the next hop RSU in its simple local routing table (see Fig. 6). Once this process is completed, then all RSUs will be able to handle such incoming data packets by modifying two specific fields in the packet reserved for both the next hop RSU id and its position, respectively. This first will help both vehicles and RSUs to discover the next hop RSU based on stored information in the packet, whereas the RSU's position will help in facilitating the precise guidance of data packets towards a specific direction based on the distance. Second, we still need to indicate how vehicles will relay communications between each other in road segments to allow connectivity between RSUs deployed at intersections, so we adopt a probabilistic broadcasting approach by taking into account the control of communication's direction. Accordingly, each item in the network can retransmit unrelated packets received from neighbor items (vehicles or RSUs) with a probability $p_i j$ based on the weighted p-persistence broadcasting scheme [39]. The broadcasting probability depends on the distance between the transmitter and the receiver. In other words, the further the nodes are located away, the higher the broadcasting probability. We exploit this option to control the communication direction towards a specific node, which is in our case the next hop RSU predefined in the shortest path. To provide precise guidance of data packets towards the convenable node, we take into account the effects of opposite traffic on the communications' direction to avoid disorientation of communications. In addition, this sure will help us to stop unneeded relays, and then help in decreasing the communication cost by an extra further amount. For this purpose, we modified the p-persistence broadcasting algorithm to take into account these assumptions. First, the p-persistence broadcasting probability is defined as follows:

$$\mathbf{p}_{i,j} = \frac{d_{i,j}}{R} \tag{8}$$

where $d_{i,j}$ is the distance between the receiver and transmitter of the packet and R is the transmission range. The successful rebroadcasting of a received packet should respect three conditions as follows (see Fig. 7):

- **Condition 1:** The vehicle should receive the packet for the first time.
- **Condition 2:** The distance between the sender and receiver should be smaller than the distance between the sender and the destination:
  Distance(receiver vehicle, next hop RSU)<Distance(sender, next hop RSU).
- **Condition 3:** The rebroadcasting probability should verify the following condition: $p_{i,j} > rand(0,1)$

If at least one of the conditions is not verified, the receiver vehicle should discard the received packet. However, under low vehicle density, the network will suffer from a lack of connectivity, and then data packets will be lost before reaching the next hop RSU. To prevent data packets lost in disconnected parts of road segments, we use a carry-and-forward mechanism to be able to deliver data packets successfully to this next hop RSU deployed at the end of the road segment. Therefore, this can be done under three different scenarios as illustrated in Fig. 8. On the other hand, if such a vehicle has already sent a request for a new shortest path to the server and it did not receive a reply, it will estimate a waiting delay to make a new tentative before arriving at the intersection by supposing that it was out of coverage. Therefore, the carry-and-forward technique will help vehicles to send successful requests when they enter into the communication range of the next hop RSU based on the following equation:

$$Distance(vehicle, next\ hop\ RSU) < R \tag{9}$$

In fact, the introduced sharing information mechanism between both vehicles and RSUs can allow a fast relaying of data via multi-hop communications as well as a low communication cost. On the other hand, we should mention that real-time path planning applications are more sensitive to latency which is a worse problem that can cause dangerous consequences, especially in emergencies. Accordingly, we think our strategy takes care of all these constraints by allowing drivers to profit from the reliability and robustness of our communication architecture.

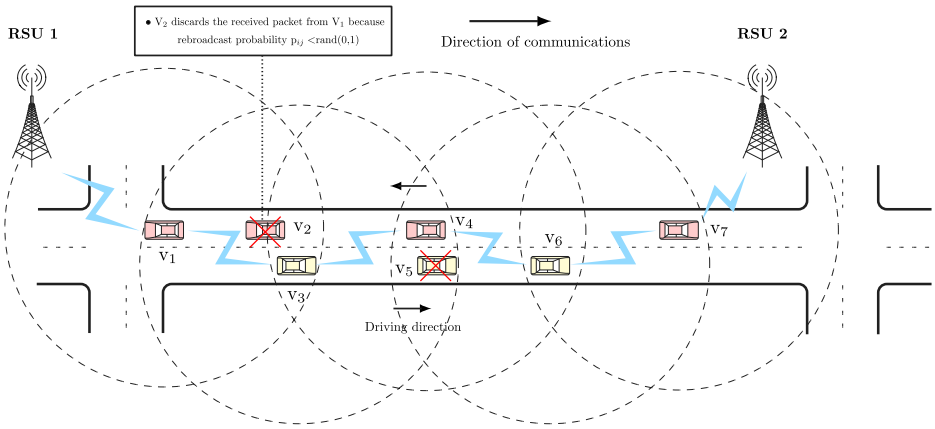


Fig. 7. Illustration of V2V communications in the case of connected road segment.

3.4. Communication cost analysis

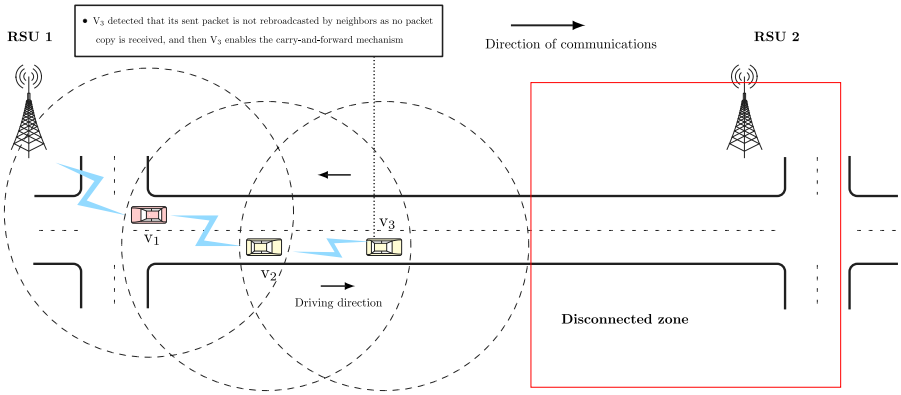
In this subsection, we describe the communication cost related to our proposed communication architecture to avoid extra communication redundancy and unneeded relaying during the collection of travel time by RSUs and in both request and reply phases, which is essential for increasing the efficiency of our path planning strategy against communication overhead. This includes improvement at different levels of communications as follows:

- **Travel time collection from vehicles (Vehicle-to-RSU)**: This operation needs only one single hop communication as each vehicle v_u broadcasts a packet when reaching the end of road segment $rs_{i,j}$. This packet includes a TTL field equals 1, an ID of corresponding road segment $rs_{i,j}$ and the travel time in road segment $rs_{i,j}$. The destination of this packet is the nearest RSU R_m located at the intersection in the neighborhood of vehicle v_u. So, the communication cost denoted by C_c is only O(1).
- **Periodic sharing of traffic information (RSUs-to-Server)**: The periodic sharing of received traffic information from vehicles by RSUs with the central server needs multi-hop communications between both vehicles and RSUs to reach the central server.

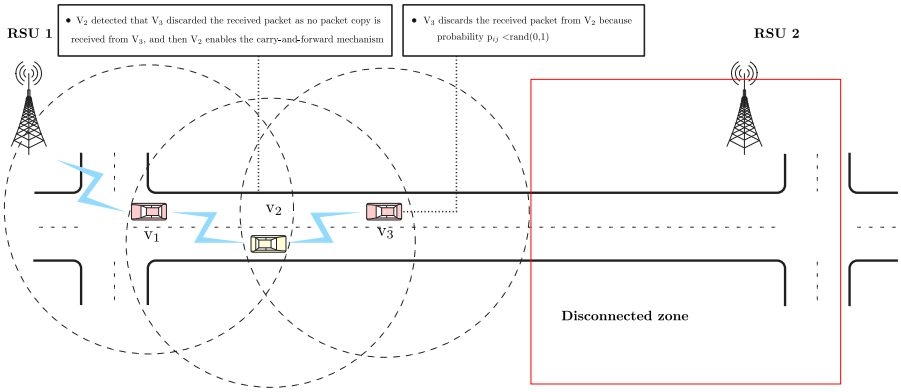
Theorem 1. *Based on our proposed shortest path-based simple routing that uses RSUs as gateways and uses a modified weighted p-persistence broadcasting between vehicles in road segments to reach the next hop RSU, the communication redundancy between items in the network can be avoided and the time complexity can be reduced as compared with broadcast-based methods.*

Proof. For forwarding communications between RSUs, the number of broadcasts depends only on the distance between the sender-receiver pair and the number of hops instead of a large diffusion of communications towards unneeded relay nodes. In this case, the communication cost will be minimized. This is because the relay nodes are selected according to the preferred communication direction based on both our simple routing and the weighted p-persistence broadcasting in road segments. So, by excluding unneeded broadcasts between both RSUs and vehicles, the overall communication cost to send a beacon is given as follows:

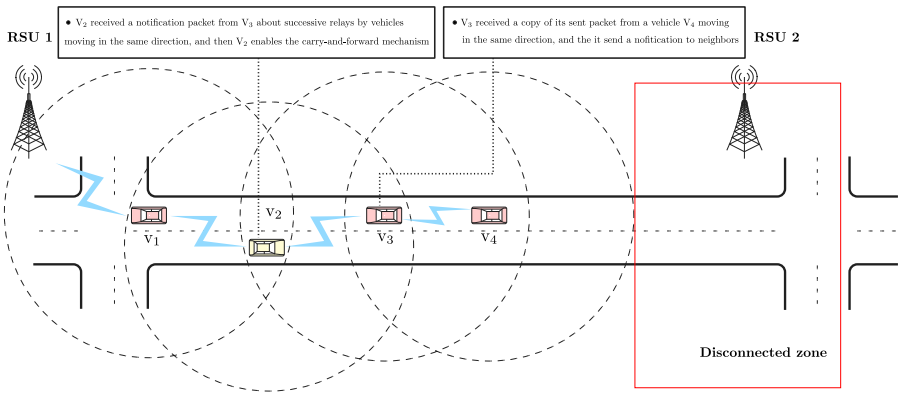
$$
\begin{aligned}
\mathbf{C}_c &= (Sent\ packet) + \big(Path\ length\ between\ RSUs - 1\big) \times \big(number\ of\ hops\ in\ each\ road\ segment\big) \\
&= 1 + \big(arg\{R_1, R_2, \ldots, R_{m'}\} - 1\big) \times \sum_{p=1}^{m'} \big(hops\ count\big)_p \\
&= 1 + \big(m' - 1\big) \times \sum_{p=1}^{m'} \big(floor\big(\frac{l_{p-1,p}}{R} + \frac{1}{2}\big) - 1\big) \\
&< O(1) + O(\log(M) \times \sum_{p=1}^{m'} \big(floor\big(\frac{l_{p-1,p}}{R} + \frac{1}{2}\big) - 1\big)\big) \\
&< O(1) + O\big(M \times \sum_{p=1}^{m'} \big(floor\big(\frac{l_{p-1,p}}{R} + \frac{1}{2}\big) - 1\big)\big) \\
&< O\big(M \times \sum_{p=1}^{m'} n_{p-1,p}\big) \quad \square
\end{aligned}
$$



(a) Disonnected road segment: Scenario 1



(b) Disonnected road segment: Scenario 2



(c) Disonnected road segment: Scenario 3

Fig. 8. Illustration of V2V communications in the case of disconnected road segment.

where M is the total number of RSUs, m' is the path's length between the source RSU and the central server (destination), $l_{p-1,p}$ is the length of road segment $rs_{p-1,p}$ and $n_{p-1,p}$ is the number of vehicles in road segment $rs_{p-1,p}$.

- **Request and reply phases (Vehicle-to-Server and Server-to-Vehicle)**: In the request and reply phases, we can easily conclude that the needed number of communications between RSUs and vehicles is twice the number of communications, which is equivalent to communication cost of one beacon transmission when such a RSU decides to share travel time information with the central server.

Theorem 2. *The need for exchanging requests and replies between vehicles and the central server is expected to increase twice the communication cost as compared with one beacon transmission during the periodic sharing of traffic information between such a RSU and the central server (see Figs. 9 and 11).*

Proof. The proof is directly based on the previous one, and therefore the overall communication cost to send a request and receive a reply is given as follows:

$$\begin{aligned}\mathbf{C}_c &= 2\times\left(1+\left(m'-1\right)\times\sum_{p=1}^{m'}\left(floor\left(\frac{l_{p-1,p}}{R}+\frac{1}{2}\right)-1\right)\right)\\ &< 2\times\left(O(1)+O(\log(M)\times\sum_{p=1}^{m'}\left(floor\left(\frac{l_{p-1,p}}{R}+\frac{1}{2}\right)-1\right)\right)\right)\\ &< 2\times O\left(M\times\sum_{p=1}^{m'}n_{p-1,p}\right) \quad \square\end{aligned}$$

Algoritmus 1: Computation of travel time on road segments.

```
/* Initialization */
E_0 = 0.1 m;
Detect = false;
/* Detect is a local boolean variable to indicate that vehicle v_u start moving on road segment rs_{i,j}. */
for each time step t do
    for each vehicle v_u in 𝕍 do
        if (Detect == false) then
            /* Calculate the distance between vehicle v_u and position Sp_{i,j} */
            if (Distance(p_u(t), Sp_{i,j}) ≤ E_0) then
                /* Here, we capture the starting time of vehicle v_u travel on road segment rs_{i,j} */
                St^u_{i,j} ←t;
                Detect = true
            end
        else if (Distance(p_u, Ep_{i,j}) ≤ E_0) then
            /* Here, we capture the arrival time of vehicle v_u on road segment rs_{i,j} */
            Et^u_{i,j} ←t;
            T_{i,j} ← (Et^u_{i,j} − St^u_{i,j}) /* (see Eq. (3)) */
            Vehicle v_u broadcast a packet containing the travel time, the corresponding road segment rs_{i,j} ID, the ID of
             destination RSU, a TTL field with value 1, and a field indicating the packet type with value 'TT'.;
            /* If this packet is received by a RSU with an id that differs from that in the received packet, then this
               packet will be dropped as it is valid only for single-hop communication. */
        end
    end
end
```

4. Performance analysis

4.1. Mobility scenario

In this section, to validate the fusibility of our path planning strategy, we use a real road network of Berlin city imported from OpenStreetMaps (OSM) based on SUMO tools as follows:

1. **Netconvert:** It converts the source file of the imported Map into one road network file supported by SUMO, which is called .net file. Moreover, several options can be used with Netconvert to provide map conversion with constraints on edges, roads, and overall road network geometry.
2. **Polyconvert:** It provides an importation of geometrical shapes (polygons) and other different sources (e.g., buildings as polygons, and so on).

We simulated different scenarios to analyze all changes and the impact related on one hand to the road network environment (i.e., traffic lights), and on the other hand to the proposed path planning parameters. The considered traffic scenarios are listed as follows:

1. **Scenario 1:** In scenario 1, we introduce a road network of 500 m × 500 that supports intersections without traffic lights, where drivers are following priority rules when arriving at any intersection.

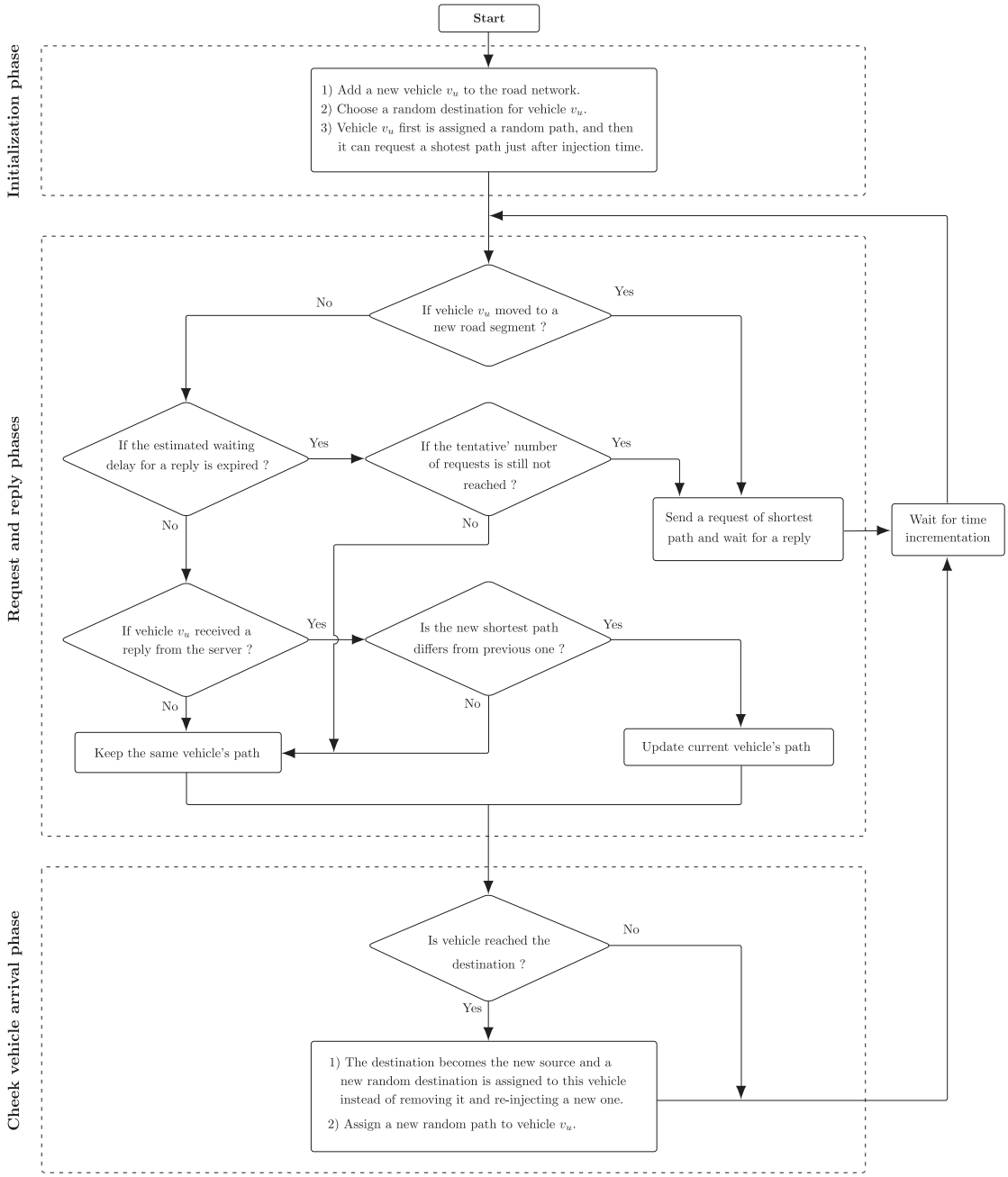


Fig. 9. Illustration of life cycle of a vehicle in the traffic system.

2. **Scenario 2:** The scenario 2 is based on the scenario 1, but all intersections are equipped with traffic lights to regulate traffic flow and manage the right of way for drivers.
3. **Scenario 3:** The scenario 3 is based on the scenario 1 and 2, where 37.5% of intersections are equipped with traffic lights, whereas other intersections are not equipped with traffic lights allowing drivers to make a decision based on priority rules instead of traffic lights (see Fig. 10).

Algoritmus 2: Computation and transfer of travel time to the Server.

```
/* Initialization */
BST ← Current time+BTI; /* BST is the beacon sending time. */
for each time step t do
    for each RSU R_m in ℝ do
        if (The RSU R_m receives a packet p_n) then
            if (The assigned packet type is 'TT') then
                Sum_tt(i, j) ← Sum_tt(i, j) + p_n.tt;
                n_{i,j} ← n_{i,j} + 1;
            else
                the RSU R_m Rebroadcasts the received packet p_n in its neighborhood to allow vehicles to complete the task
                  via V2V communications based on both weighted p-persistence broadcasting and carry-and-forward
                  mechanism in the case of disconnected parts of the road segment.
            end
        end
        if (BST == t) then
            The RSU R_m sends a beacon containing the average travel time for each nearest road segment rs_{i,j}, the id of
              road segment rs_{i,j}, and packet type 'TTU', where the average travel time equals to (Sum_tt(i, j)/n_{i,j});
            BST ← BST+BTI;
            Start a new time window for the computation of average travel time.
        else
            Moving the time window with one step time; /* See Fig. 5 for more details about the sliding time window. */
        end
    end
    if (The server receives a beacon with type 'TTU') then
        The server updates the travel time in the database based on the road segments' IDs stored in the header of each
          received beacon.
    end
end
```

First, we inject vehicles periodically into the traffic system, where each injected vehicle is assigned a random pair of origin–destination and a random path to the destination based on the distance. Each path is defined as a sequence of turning movements (i.e., right, left, straight ...). The turning movements determine if the vehicle will continue moving in a straight direction when arriving at such an intersection or if it will make a specific turn (right, left, or U-turn). Next, the injected vehicle can request a new shortest path and then update the previous path (see Fig. 9). However, at the beginning of the simulation, the server still had not received information about the traffic state. Accordingly, it provides the shortest paths based on the distance as travel time information is still absent in the database DB_{ser}.

To keep a specific vehicle density in the road network, each vehicle that reached its destination will be assigned a new random destination, while the previous destination will become the new origin and verse versa. Accordingly, three vehicle densities are considered as follows:

- **Low vehicle density**: Is defined by 50 vehicles as the maximum number of vehicles which are allowed to use the road network. When a vehicle reaches the destination, it will be assigned a new random destination instead of injecting a new one.
- **Medium vehicle density**: Is defined by 100 vehicles as the maximum number of vehicles which are allowed to use the road network.
- **High vehicle density**: Is defined by 150 vehicles as the maximum number of vehicles which are allowed to use the road network.

4.2. Simulation environment and network model

Different simulation environments were performed to evaluate the performance of our path planning strategy. First, SUMO (Simulation of Urban Mobility) a microscopic road traffic simulation, is used to simulate the traffic dynamically on a real map. Second, VENS is a network simulator capable of simulation VANETS based on different modules specifically used to enable communication in a vehicular environment. This simulator is used with OMNET++ (Network simulator) and SUMO. We evaluated our path planning strategy in VEINS based on IEEE 802.11 p/WAVE standard under different vehicle densities. In our scenario, we analyzed two main performance indicators, the end-to-end delay and the average travel time.

Algoritmus 3: Request phase

```
for each vehicle v_u in 𝕍 do
    I^id_cur−s = GetCurSourInt(Cur−Road−ID);
    /* GetCurSourInt(Cur-Road-ID) is a function called to search for a source intersection. */
    if ((I^id_cur−s ≠ I^id_prev−s) or (I^id_cur−s == I^id_prev−s and The attempts number of requests sending < A predefined attempts number))
      then
        if (The vehicle v_u is in waiting mode for a reply) then
            if (Estimated waiting delay for a reply < Time threshold) then
                The vehicle v_u avoids sending a new request of shortest path;
            else if (The attempts number of requests sending < A predefined attempts number) then
                Vehicle v_u broadcasts a request packet to the server containing {vehicle's id, I^id_cur−s, I^id_dest, R_sour, R_dest, and
                  R_next−hop} ;
                Increment the attempts number of requests sending;
                Vehicle v_u starts waiting for a reply from the server;
            end
        else if (The attempts number of requests sending < A predefined attempts number) then
            Vehicle v_u broadcasts a request packet to the server containing {vehicle's id, I^id_cur−s, I^id_dest, R_sour, R_dest, and
              R_next−hop} ;
            Increment the attempts number of requests sending;
            Vehicle v_u starts waiting for a reply from the server;
        end
        if (I^id_cur−s ≠ I^id_prev−s) then
            I^id_prev−s ← I^id_cur−s;
        end
    end
end
```

Algoritmus 4: Reply phase.

```
for each vehicle v_u in 𝕍 do
    if (A reply is received by vehicle v_u) then
        if (The vehicle v_u is the destination of sent reply) then
            if (The received shortest path differs from the previous one) then
                The vehicle v_u updates its path with the newly received shortest path.
            else
                The vehicle v_u keeps using the same vehicle's shortest path.
            end
        else
            The vehicle v_u rebroadcasts the rely based on the weighted p-persistence broadcasting scheme.
        end
    end
end
```

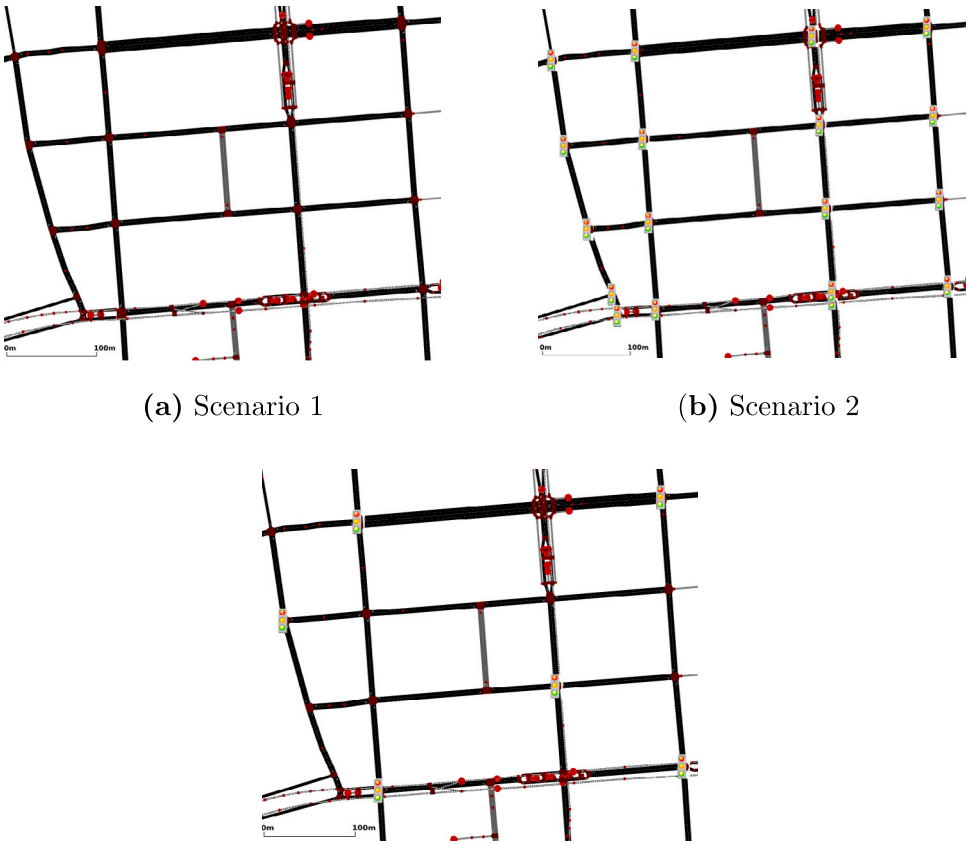
4.3. Indicators of system performance

In this section, we introduce a set of metrics to evaluate the performance of our path planning strategy, including travel time, the number of travels, and the mean speed. The performance metrics in this work are described in detail as follows:

- **Mean speed:** We define the mean speed as of average of vehicles' speed during a specific time interval T'. The mean speed can be calculated as:

$$\textbf{Mean speed} = \frac{1}{T' \times N} \sum_{t=1}^{t=T'} \sum_{u=1}^{u=N} s_u(t) \tag{10}$$

  where N is the number of vehicles in the road network and the value of T' in our case is equal to 1 min.
- **Travel time:** Is defined as the average travel time for a set of vehicles traveling between a specific source S and a destination D, where the travel time for each vehicle is the time difference resulting between the start time of travel ($t^{(i)}_{start}$) when such a vehicle u starts moving from a source location and the arrival time ($t^{(i)}_{arrival}$) when the vehicle u reaches the destination



(a) Scenario 1 (b) Scenario 2

(c) Scenario 3

Fig. 10. Illustration of different road network scenarios considered in our simulation: (a) Represents scenario 1, (b) Represents scenario 2, and (c) Represents scenario 3.

Table 3
Simulation parameters.

Parameter	Value
Roadmap size	500 m × 500 m
Simulation time	4000 s
Network simulator	Veins 4.4
Size of message	500 bytes
MAC/PHY	IEEE 802.11 p/WAVE
Frequency	5.89 GHz
Channel bandwidth	10 MHz
Bitrate rate	18 Mb/s
Communication range	100 m
TTL of packet	32
Number of RSUs	16
Obstacles	Without
Mobility simulator	SUMO 0.25.0
The model used for car following	carFollowModel
Lane change model	LC2013

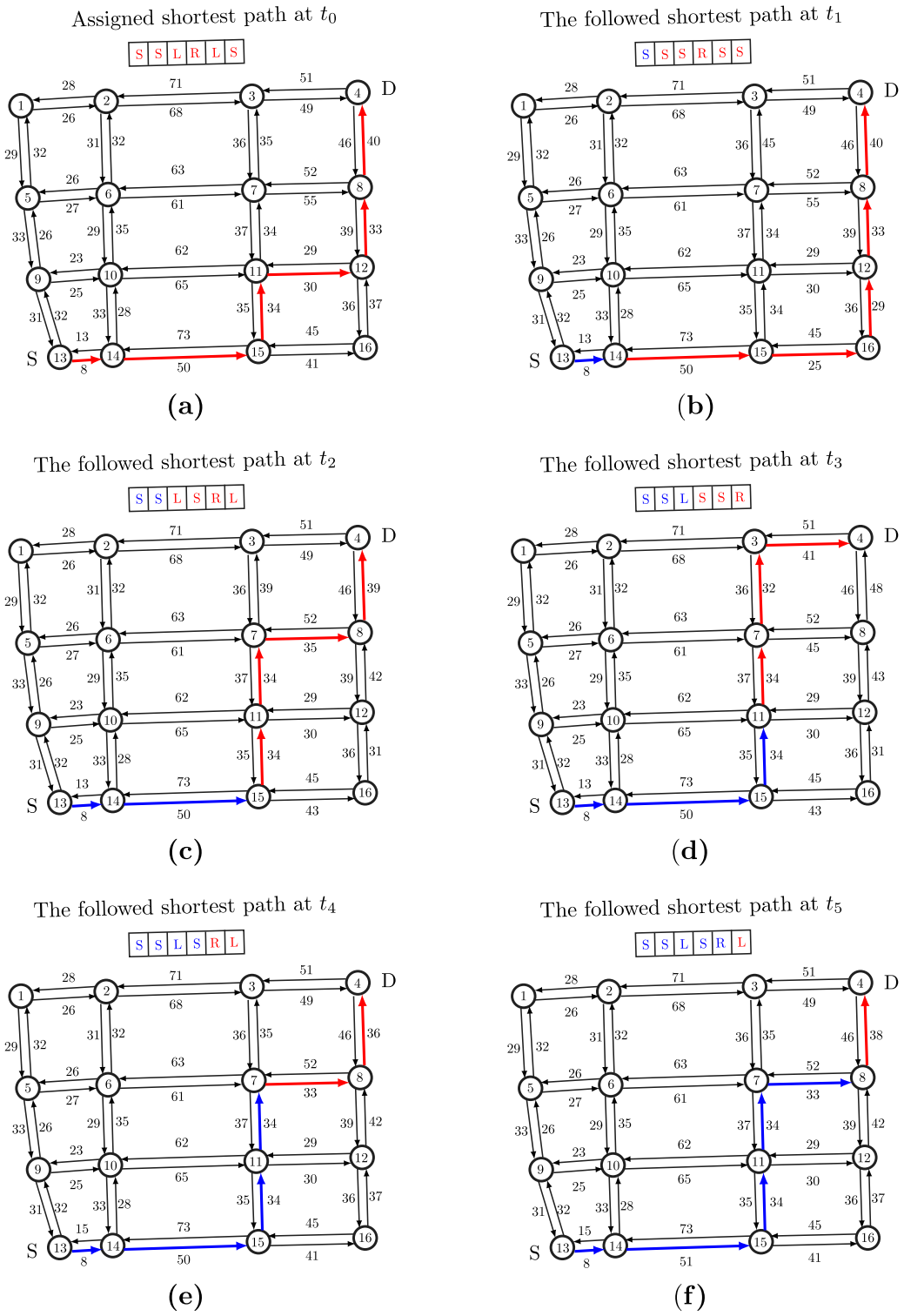


Fig. 11. Illustration example of vehicle's traveling path dynamics versus traffic state.

location. Here, we consider the possibility of redundant traveling between S and D for each vehicle. Indeed, the travel time can be defined as follows:

$$\textbf{Travel time} = \frac{\sum_{i=1}^{i=M'} (t_{arrival}^{(i)} - t_{start}^{(i)})}{M'} \tag{11}$$

where i is the id of travel and M' is the total number of travels between S and D (S = 1 and D = 16) (resp. S = 16 and D = 1).

- **Total number of travels:** We define the total number of travels as the number of times that vehicles traveled between a specific source and destination pair. Each travel is recorded based on start time ($t_{start}^{(u)}$) when such a vehicle u starts moving from its assigned source location and the arrival time ($t_{arrival}^{(u)}$) when the vehicle reaches its destination.
- **Packet delivery ratio:** The packet delivery ratio is defined as the total number of packets successfully received by the destination divided by the total number of data packets sent from the source nodes.
- **Packets volume:** The packets volume includes the overall sent packets during a period of time.
- **Communication redundancy:** The communication redundancy is the overall number of retransmissions of the same packets during a specific period of time.
- **Latency:** The latency is the delay between the first time a packet is sent from the source and the time the packet is received by the destination.

To efficiently monitor both the travel time and the total number of travels, we select five vehicles at the beginning of the simulation as reference vehicles and force them to travel between a specific source–destination pair. Accordingly, we selected the longest source–destination pair which is (S = 1 and D = 16) (resp. S = 16 and D = 1) to make sure that vehicles trajectories are covering different regions of the traffic scenario, and therefore the statistics will reflect the traffic state correctly. When these vehicles reach their destinations, the source becomes the new destination and the previous destination becomes the new source, and verse versa.

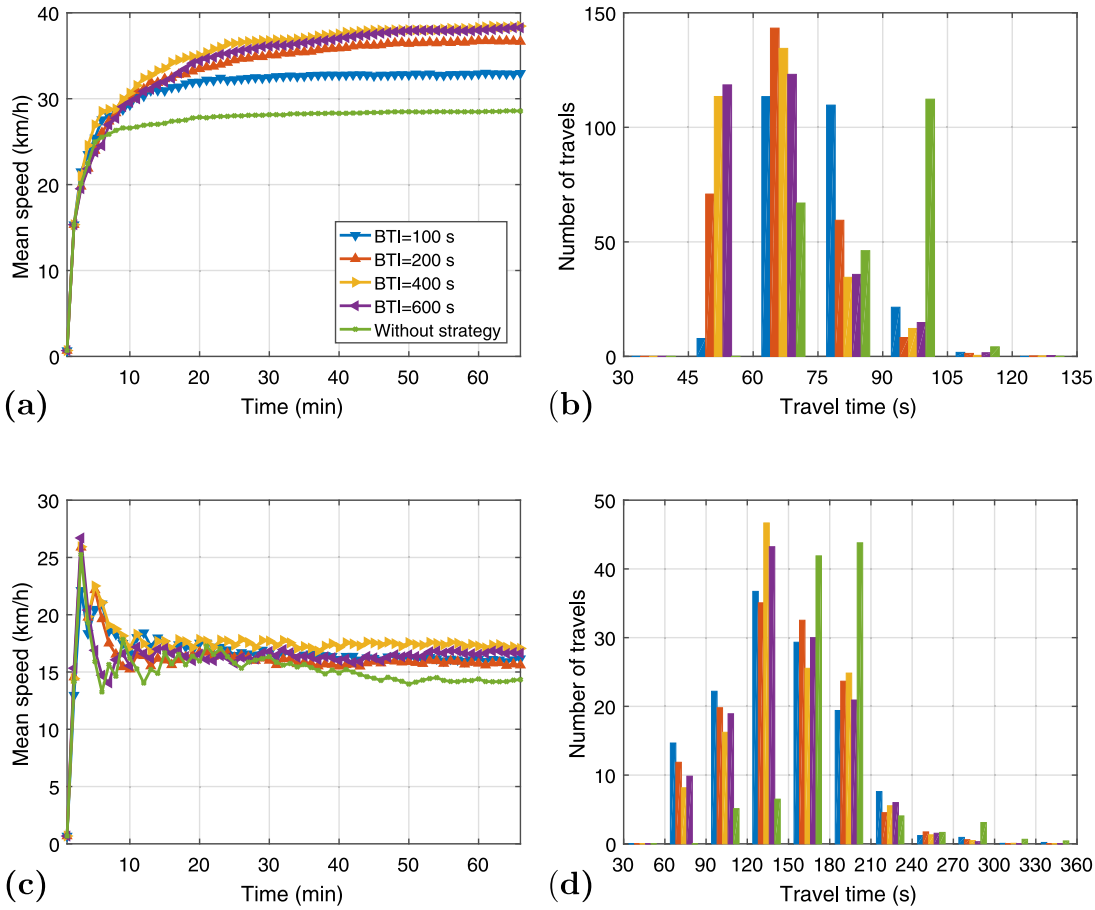


Fig. 12. Evaluation of the path planning strategy with and without traffic lights under low vehicle density. (a) and (b) represent the mean speed and travel time without traffic lights. (c) and (d) represent the mean speed and travel time with traffic lights. BTI denotes beacons time interval and WS denotes without a strategy.

Table 4
Comparison in terms of total number of travels under low vehicle density by considering two scenarios: with traffic lights and without traffic lights.

Density of vehicles	Scenario without traffic lights						
	BTI=	100 s	200 s	400 s	600 s	Sum	Without strategy
Low		254.24	283.38	295.33	294.27	1127.22	229.62
	Scenario with traffic lights						
	BTI=	100 s	200 s	400 s	600 s	Sum	Without strategy
Low		132.33	129.8	128.85	130.46	521.44	108.01

5. Simulation results and discussions

In this section, to validate the feasibility and the effectiveness of our path planning strategy, we adopt different traffic scenarios (see Fig. 10) and vehicle density (low, medium, and high). To that end, a set of performance metrics is considered to depict the system performance from different sides, including the mean speed, travel time, and the total number of travels. Next, to prove the effectiveness of the communication architecture in reducing the communication cost while providing information sharing between components in the network, we evaluate the performance of communication architecture in terms of packets volume. We run our path planning strategy using both Veins 4.3 and SUMO 0.25.0 simulators. The parameters used in the simulation for both mobility and network model are summarized in Table 3.

5.1. Evaluation of the path planning strategy

In Fig. 12, we plot both the mean speed and travel time under low vehicle density for different traffic scenarios (scenarios 1 and 2) and BTIs.

In the first scenario: under low vehicle density, Fig. 12a, b show that the path planning strategy significantly improves both the mean speed and travel time. Furthermore, when increasing the BTI, the mean speed increases, whereas the travel time decreases as a longer BTI can lead to a sufficient collection of traffic information in terms of travel time, especially under low vehicle density. This can drive us to think that the collection of travel time information in road segments provided by RSUs in the case of long BTIs is expected to be the most optimal for low vehicle densities. Therefore, this is expected to help us make more accurate decisions on the traffic state in such a road segment. On the other hand, Table 4 shows that the obtained results in terms of the total number of travels match very well with traffic patterns observed in obtained results in terms of both mean speed and travel time, where the most higher total number of travels is observed when adopting long beacon time interval. Differently from short beacon time intervals, long beacon time intervals allow vehicles to move very fast and travel many times between the observed origin-destination pair by following the best shortest paths.

In the second scenario: the path planning strategy provides a limited improvement of both the mean speed and travel time (see Fig. 12c, d). In addition, when increasing the BTI, the mean speed still increases but slightly, whereas the travel time is still decreasing significantly regardless of the presence of traffic lights. By comparing both Fig. 12a, b and c, d, we can observe that both the mean speed and travel time in the case of scenario 1 (road network with unsignaled intersections) outperform more significantly those observed in the case of scenario 2 (road network with signalized intersections). In fact, it is well known that intersections equipped with traffic lights affect the traffic state when they have compared with roundabouts or unsignaled intersections as both these last follow priority rules instead of traffic lights, and therefore this is expected to avoid extra waiting time for drivers and leads to an improvement in the traffic flow. In addition, Table 4 shows that the obtained results in terms of the total number of travels in the case of scenario 1 outperform those observed in the case of scenario 2 with a percentage of about 53.74% (with 521.44 travels for scenario 2, which represents 46.26% only against 1127.22 travels for scenario 1), and these findings match very well with our previous comments. Also, in the case of scenario 2, it is observed that the BTI has less effect on the total number of travels as vehicles' speed is constrained by traffic lights at intersections.

In Fig. 13, we plot both the mean speed and travel time under medium vehicle density for different traffic scenarios (scenarios 1 and 2) and BTIs.

In the first scenario: under medium vehicle density, Fig. 13a, b show that the path planning strategy still significantly improves both the mean speed and travel time. Similar to the results for scenario 1 with low vehicle density, Fig. 13a, b reveal that the high increase in BTI significantly improves both the mean speed and travel time. Differently from results in Fig. 12a, b, the obtained results under medium vehicle density show an overall decrease in the mean speed of about 6.25% for all BTIs as compared to previous results in the case of scenario 1 (see Fig. 13a, b). Moreover, under increased BTI, a high number of travels are shown in the interval [50,80] min (see Fig. 13b), whereas under short BTI, the number of travels is heavily decreased. Based on the further comparison with the results from Fig. 12a, b, it is clear that the decrease in both mean speed and the increase in the travel time are related mainly to an increase in the vehicle density, where vehicles' speed becomes more constrained by traffic state, and therefore this means a direct increase in the travel time and a decrease in vehicle speed. This is also confirmed by the simulation results illustrated in Tables 4 and 5, where the sum of the total number of travels for all BTIs is clearly decreased.

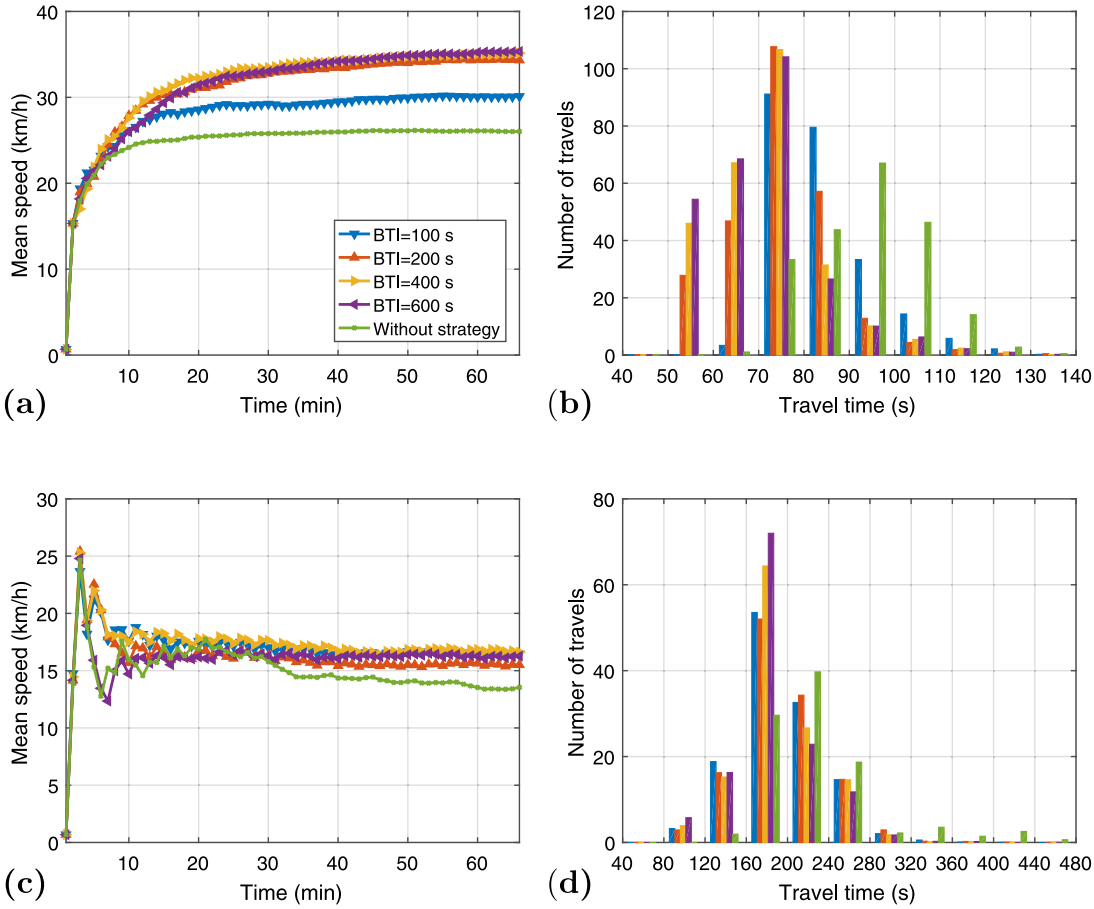


Fig. 13. Evaluation of the path planning strategy with and without traffic lights under medium vehicle density. (a) and (b) represent the mean speed and travel time without traffic lights. (c) and (d) represent the mean speed and travel time with traffic lights.

Table 5
Comparison in terms of total number of travels under medium vehicle density by considering two scenarios: with traffic lights and without traffic lights.

Density of vehicles	Scenario without traffic lights						
	BTI=	100 s	200 s	400 s	600 s	Sum	Without strategy
Medium		230.12	259.9	270.77	273.61	1034.4	209.06
	Scenario with traffic lights						
	BTI=	100 s	200 s	400 s	600 s	Sum	Without strategy
Medium		125.48	123.43	126.73	130.75	506.39	95.8

In the second scenario: the path planning strategy is still providing a limited improvement of both the mean speed and travel time (see Fig. 13c, d). However, this improvement appears to be limited when it is compared to results in Fig. 13c, d. In addition, when increasing the BTI, the mean speed is still increasing slightly, while the travel time is still decreasing significantly regardless of the presence of traffic lights, especially for long BTIs. Moreover, the results from Table 5 show that the sum of the total number of travels in the case of scenario 1 for all BTIs is decreased by 8.23% as compared with the previous results for the same scenario under low vehicle density (i.e., 1034.4 travels against 1127.22 travels). Also, we observed that the sum of the total number of travels in the case of scenario 2 is slightly decreased by 2.89% as compared with the previous results for the same scenario under low vehicle density (i.e., 506.39 travels against 521.44 travels). On the other hand, we observe that in both scenarios, the long BTIs show better results in terms of the total number of travels.

In Fig. 14, we plot both the mean speed and travel time under high vehicle density for different traffic scenarios (scenarios 1 and 2) and BTIs.

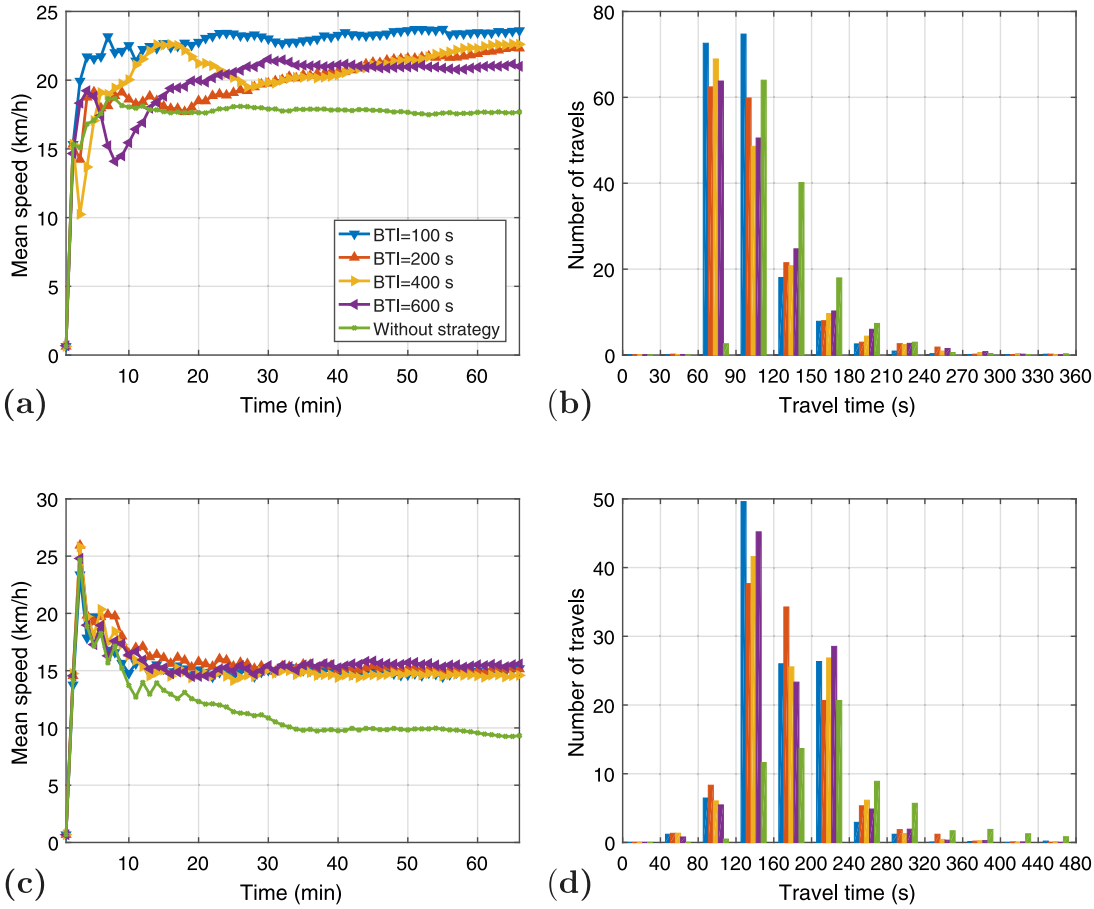


Fig. 14. Evaluation of the path planning strategy with and without traffic lights under higher vehicle density. (a) and (b) represent the mean speed and travel time without traffic lights. (c) and (d) represent the mean speed and travel time with traffic lights.

Table 6
Comparison in terms of total number of travels high low vehicle density by considering two scenarios: with traffic lights and without traffic lights.

Density of vehicles	Scenario without traffic lights						
	BTI=	100 s	200 s	400 s	600 s	Sum	Without strategy
High		177.6	160.01	156.87	160.73	655.21	136.5
	Scenario with traffic lights						
	BTI=	100 s	200 s	400 s	600 s	Sum	Without strategy
High		114.13	110.93	109.65	110.8	445.51	66.77

In the first scenario: under high vehicle density, Fig. 14a, b show that the path planning strategy can still significantly improve both the mean speed and travel time, even if vehicle density is increased. Similarly, in Fig. 14a, b the obtained results in the case of scenario 1 under high vehicle density show that the high increase in the beacon time interval can still significantly improves both the mean speed and travel time. Surprisingly, both mean speed and travel can get improved when decreasing the BTI, where the best results are observed in the case of 100 s. This is expected to be due to the fast transition in the traffic state, and therefore short BTIs can become more convenient to inform the server about new changes in the road network. However, we observe an overall decrease in the mean speed of $\approx 40\%$ as compared with that in the case of low vehicle density. Moreover, a heavy decrease in the frequency of travels number is observed in the interval [0,60] min, which means that the traffic state is significantly affected by the increase in the vehicle density, and therefore both the vehicle speed and travel time are affected by the increase in vehicle density. Consequently, it appears that long BTIs are more convenient to use under high vehicle density and congested scenarios, whereas short BTIs are more convenient to use under low vehicle density. This means that the choice of BTI can differ depending on the traffic state in each specific region.

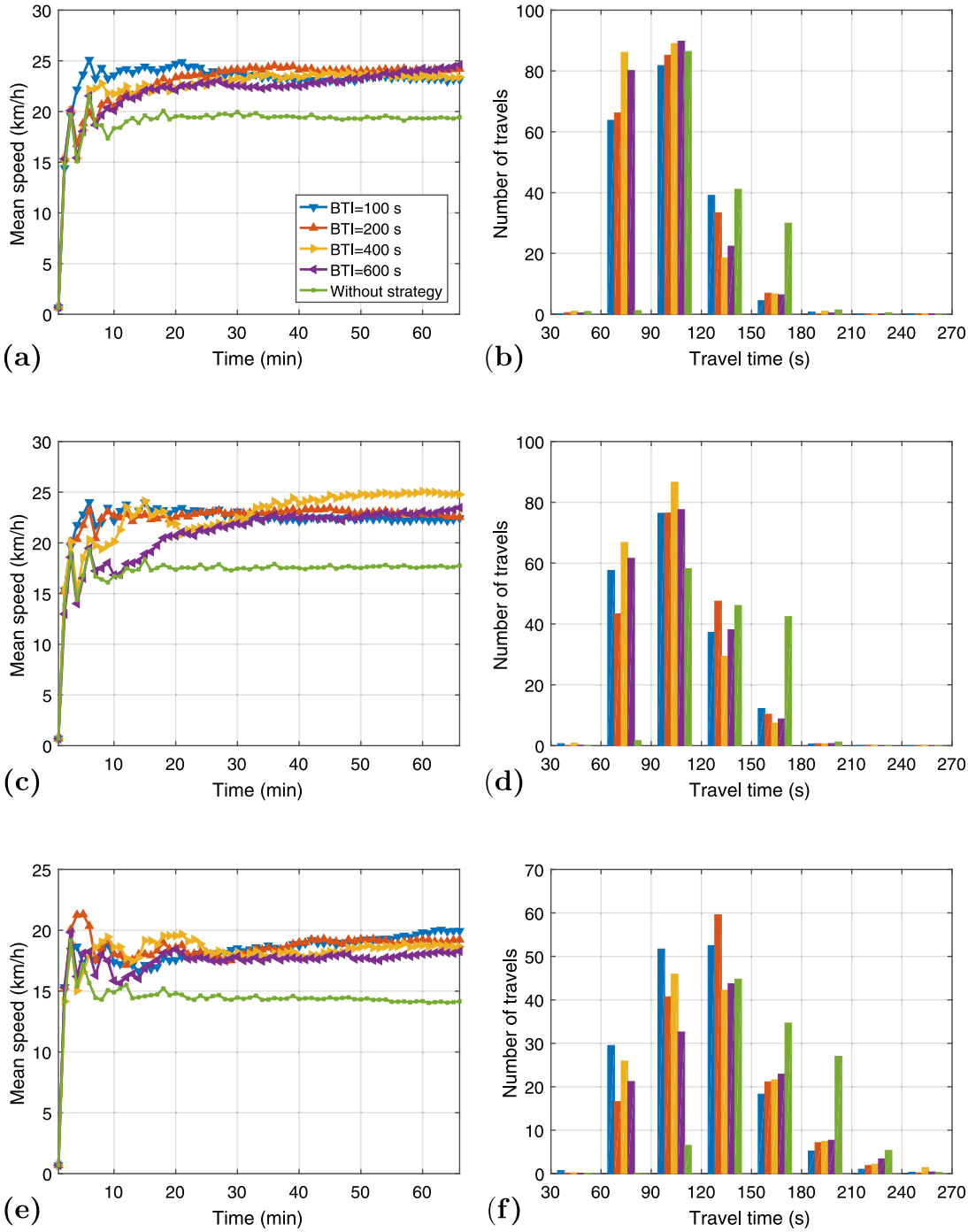


Fig. 15. Evaluation of the path planning strategy in the case of scenario 4 under different vehicle densities (low, medium, and high). (a) and (b) under low density, (c) and (d) under medium density, and (e) and (f) under high density.

In the second scenario: the path planning strategy shows a remarkable improvement in both the mean speed and travel time (see Fig. 14c, d). Different from previous results under low and medium vehicle density, under high vehicle density, the results show that our path planning can improve the mean speed until about 33%. We observe also that when our path planning strategy is not used, the transition in the mean speed appears to be too fast. But, under both low and medium vehicle density, the transition in the mean speed was very slow. This means that under high vehicle density, the change in the traffic state is very fast, therefore this can contribute to the appearance of congested traffic scenarios.

Table 7
Comparison in terms of total number of travels under different vehicle density (low, medium and high) by considering two scenarios: with traffic lights and without traffic lights.

Density of vehicles	Scenario with and without traffic lights					
	BTI=	100 s	200 s	400 s	600 s	Sum	Without strategy
Low		190.08	192.33	202.36	199.6	784.37	161.51
Medium		184.73	178.4	191.6	186.9	741.63	149.56
High		159.2	147.29	146.8	131.9	585.19	118.66

On the other hand, the results from Table 6 show that the sum of the total number of travels in the case of scenario 1 for all BTIs is decreased by 36.66% as compared with previous results under medium vehicle density. Moreover, the sum of the total number of travels in the case of scenario 2 is decreased by 12.02% as compared with previous results under medium vehicle density. Differently from previous results, we observe that in both scenarios, the low BTIs show better results in terms of the total number of travels. This confirms that the results in terms of the sum of the total number of travels match very well with the traffic state illustrated in terms of both mean speed and travel time in Fig. 14. In Fig. 15, we plot both the mean speed and travel time in the case of scenario 3 under varying vehicle density (low, medium, and high) and varying BTIs.

Fig. 15a, b show that under low vehicle density, the path planning strategy significantly improves both the mean speed and travel time, but the results follow a different pattern as compared with results for scenario 1 under the same vehicle density. Moreover, we observe an overall decrease of the mean speed of $\approx 34.21\%$ as compared with scenario 1 for the same vehicle density. In addition, Fig. 15b shows a remarkable increase in the travel time, where a huge decrease in the number of the travels is observed in the interval [0,90] min. This is because, in the third scenario, 37.5% of intersections are equipped with traffic lights, which have a worse impact on both mean speed and travel time. Accordingly, the results from Table 7 confirmed this where the total number of travels is decreased from 1127.22 to 784.37 as compared to previous results under low vehicle density for scenario 1. This means a reduction of 30.42% of the number of travels, but for scenario 2 under the same vehicle density, this is considered as an improvement of 33.52%. On the other hand, from Fig. 15a, b, it can be also observed that the long BTIs are still to be the most convenient BTI for the periodic exchange of traffic information between RSUs and the server under low vehicle density. Here, it is expected that short BTIs may cause some perturbations to the traffic system as short BTI will lead to a fast change in the traffic state. This is caused by a high demand for shortest paths under a short time interval. In other words, this is known as a new congestion generation caused by a large number of vehicles that are assigned new shortest paths with common road segments characterized by short travel time.

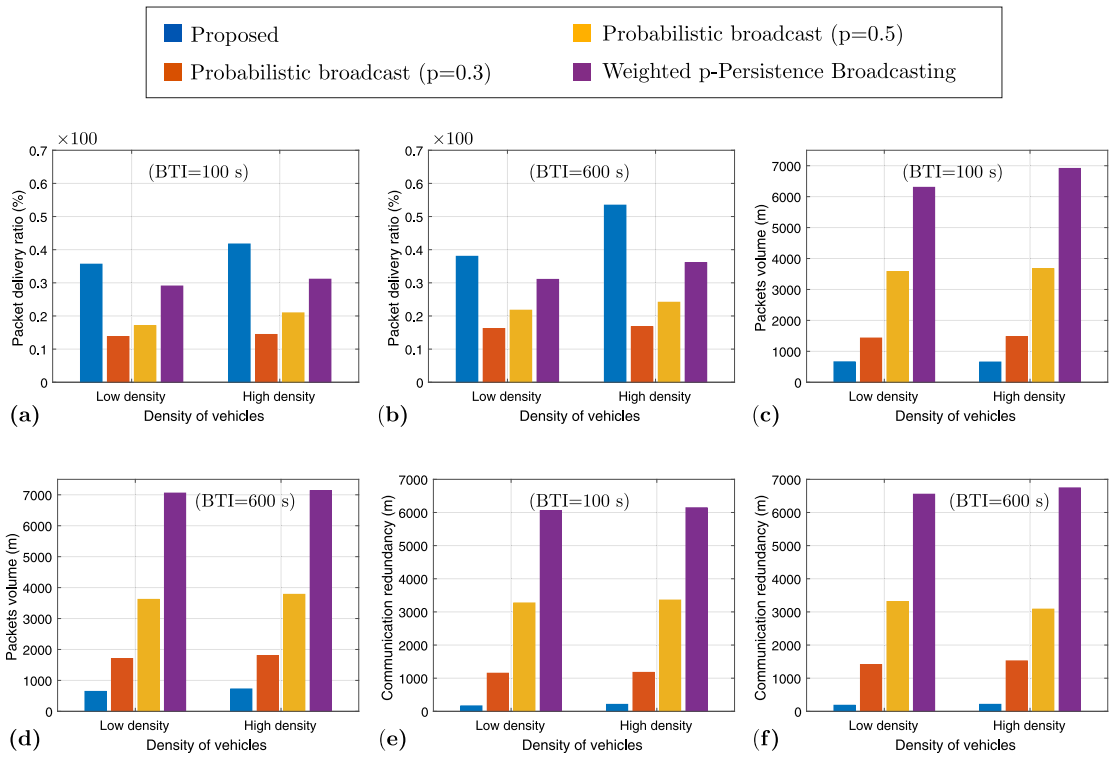


Fig. 16. Evaluation of the communication cost for the path planning strategy under low and high vehicle density for different beacon time intervals (BTI = 100 s and BTI = 600 s): (a) and (b) Packet delivery ratio (PDR), (c)–(d) Packets volume (m), (e)–(f) Communication redundancy (m) and m means minute.

Fig. 15c–d show that under medium vehicle density, the path planning strategy still improves both the mean speed and travel time, regardless of a limited decrease in the mean speed for all BTIs except for BTI = 400 s which leads to a remarkable increase of mean speed in the time interval [30,65] min. This increase in the mean speed in the case of (BTI = 400 s) is because long BTIs are the most convenient in the case of increased vehicle density. However, we observe that the travel time is increased by about 5.45% as compared with previous results under low vehicle density. From Table 7, we observe that the total number of travels decreased from 784.37 to 741.63 as compared to results under low vehicle density for the same scenario. This means a reduction of 5.45% of travels, but for scenario 1 under the same vehicle density, the total number of travels is decreased by 28.30%, whereas for scenario 2, this is considered as an improvement of 31.72%.

Fig. 15e–f show that under high vehicle density, both the mean speed and travel time are improved significantly when path planning is used. Surprisingly, the shortest BTI (100 s) is considered the best one that can help in improving both mean speed and travel time, whereas the long BTIs (> 100 s) show the lowest performance. This confirms again that the long BTIs is the best choice under higher traffic density. From Table 7, we observe that the total number of travels decreased from 741.63 to 585.19 as compared to results under medium vehicle density for the same scenario. This means a reduction of 21.09% of travels number. But, when comparing the obtained results with those in scenario 1 under the same vehicle density, we observe that the total number of travels is decreased by 10.69%, whereas the comparison with obtained results in the case of scenario 2 under the same vehicle density shows an improvement of 31.52%.

5.2. Evaluation of communication cost

In this subsection, we analyze our proposed communication architecture according to communication cost based on several performance metrics, including Packet delivery ratio (PDR), Packets volume (m), Communication redundancy (m), and latency per each type of packet (i.e., requests, replies, and beacons). We used two different vehicle densities (low and high) and two different BTIs (BTI = 100 s and BTI = 600 s), while the adopted traffic scenario is scenario 3. Furthermore, we provide a comparison with other broadcasting algorithms to show how much our proposed communication architecture can decrease communication cost and improve the reliability of the path-planning strategy. The comparison includes probabilistic broadcast (p = 0.3), Probabilistic broadcast (p = 0.5), and Weighted p-Persistence Broadcasting.

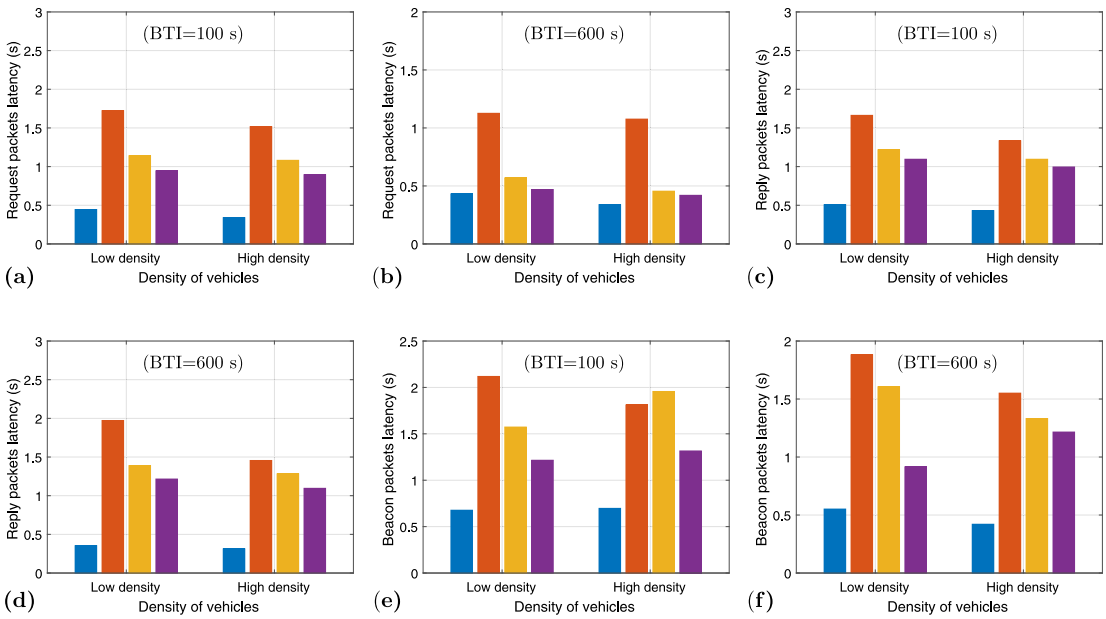


Fig. 17. Evaluation of the communication cost for the path planning strategy under low and high vehicle density for different beacon time intervals (BTI = 100 s and BTI = 600 s): (a) and (b) Request packets latency (s), (c)–(d) Reply packets latency (s), (e)–(f) Beacon packets latency (s) and s means second.

Fig. 16a–b show that under low vehicle density and low BTIs, the proposed communication strategy outperforms the other methods. Moreover, the obtained results show that the other methods suffer from a low reachability problem. Under high vehicle density, our strategy significantly outperforms the other methods due to an increase in the number of relay nodes in road segments, which helps in improving reachability. When increasing vehicle density, a considerable improvement in packet delivery ratio is observed in the case of our strategy, whereas a slight improvement is observed for the other methods. Under both long BTIs and high vehicle density, we observed that the packet delivery ratio in the case of our strategy gets improved significantly due to a decrease in the communication cast under long BTIs (fewer beacons exchange in the network).

In Fig. 16c–d, we can see that our communication strategy shows a reduced packet volume under both low and high vehicle density and short BTI, whereas the other methods show a higher packets volume, especially for the Weighted p-Persistance Broadcasting. This is mainly due to a high retransmission probability, which can reach until 1 in the case of the furthest vehicle/RSU of the sender. However, this is not sufficient to guarantee that such a packet can reach the destination in the absence of any control of communication orientation toward the destination (routing). Under, higher vehicle density, the packet volume increases significantly, however, our strategy is still showing the best performance in terms of packets volume with a reduction of 50% as compared with the other methods.

In Fig. 16e–f show that under low vehicle density and small BIT, the communication redundancy is highly increased in the case of broadcasting methods due to frequent broadcasting of the same packet by different vehicles and RSUs. Similarly, when increasing the vehicle density, the same behavior is observed as the number of broadcast attempts is increased. Under long BTIs, the communication redundancy increases significantly for all broadcasting methods, but less than that observed in the case of short BTI. In the contrast, our strategy is noticeably less affected by both high vehicle density and long BTI length.

In Fig. 17, we observe that under higher vehicle density and long BTIs, our strategy outperforms the other methods and shows the best latency delay, while in the case of higher vehicle density and short BTIs, the latency is slightly decreased. This means that, under the short BTIs, the latency is affected by the high rate of packets volume circulating in the network due to the high beaconing rate adopted by RSUs. However, our strategy still significantly outperforms the other methods, which confirms the reliability of our strategy to be a suitable and confident communication solution for path-planning applications and other real-time applications. Moreover, the observed shift in performance observed when our path planning strategy is compared with the other methods reflects the communication condition in the network over which nodes encounter difficulties to relay unrelated packets successfully to such a destination due to high communication cost and the absence of an oriented relying mechanism. Consequently, this may expose the traffic system to huge perturbations and expose the drivers to a dangerous situation on the road.

6. Conclusion

In this paper, we proposed a real-time path planning strategy to reduce vehicle traveling time based on a distributed and cooperative information sharing mechanism for avoiding traffic congestion by taking into account the communication overhead problem caused by redundant communications in the network. Accordingly, an effective cooperative exchange of communications is provided in different phases, including traffic information sharing phase, request phase, and reply phase. In all these phases, the Real-time path planning strategy is capable of providing multi-hop communications by avoiding both communication redundancy and reducing communication cost from $O(m \times n)$ to $O(\log(m+n))$ and less. Accordingly, varying methods are proposed to optimize the communication architecture to be able to control efficiently communication overhead. First, the assigned roles for each item in the network are revised to avoid redundant communications. Next, we proposed a simple shortest path-based routing that uses RSUs as gateways to relay communications towards a specific destination instead of using broadcasts. Finally, to avoid disconnectivity problems in some road segments between RSUs and communications redundancy, a weighted p-persistence broadcasting method is proposed to control both communication's direction and avoid unneeded communications relays, which can avoid a huge communication cost in the network. We evaluated the proposed path planning strategy according to various parameters, including traffic scenarios, vehicle density, and BTI. Accordingly, several performance metrics are adopted to reveal the reliability of our strategy, including mean speed, travel time, and the total number of travels, while packet delivery ratio, packets volume, communication redundancy, and latency are provided to evaluate the communication cost of our proposed communication architecture. The simulation results show that our strategy could improve significantly both the mean speed of vehicles and the travel time. Furthermore, we found that the use of short BTIs are more convenient under increased traffic density, whereas long BTIs are convenient under low traffic density. Moreover, the the proposed communication architecture can avoid redundant communication and reduce significantly the communication cost. In future work, the follow-up of this work could drive us to assess our path planning strategy under more challenging traffic scenarios on the road such as accidents, gridlock and so on. In addition, we also intend to incorporate new enhancements to avoid some phenomena like the new congestion generation.

Data availability

No data was used for the research described in the article.

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